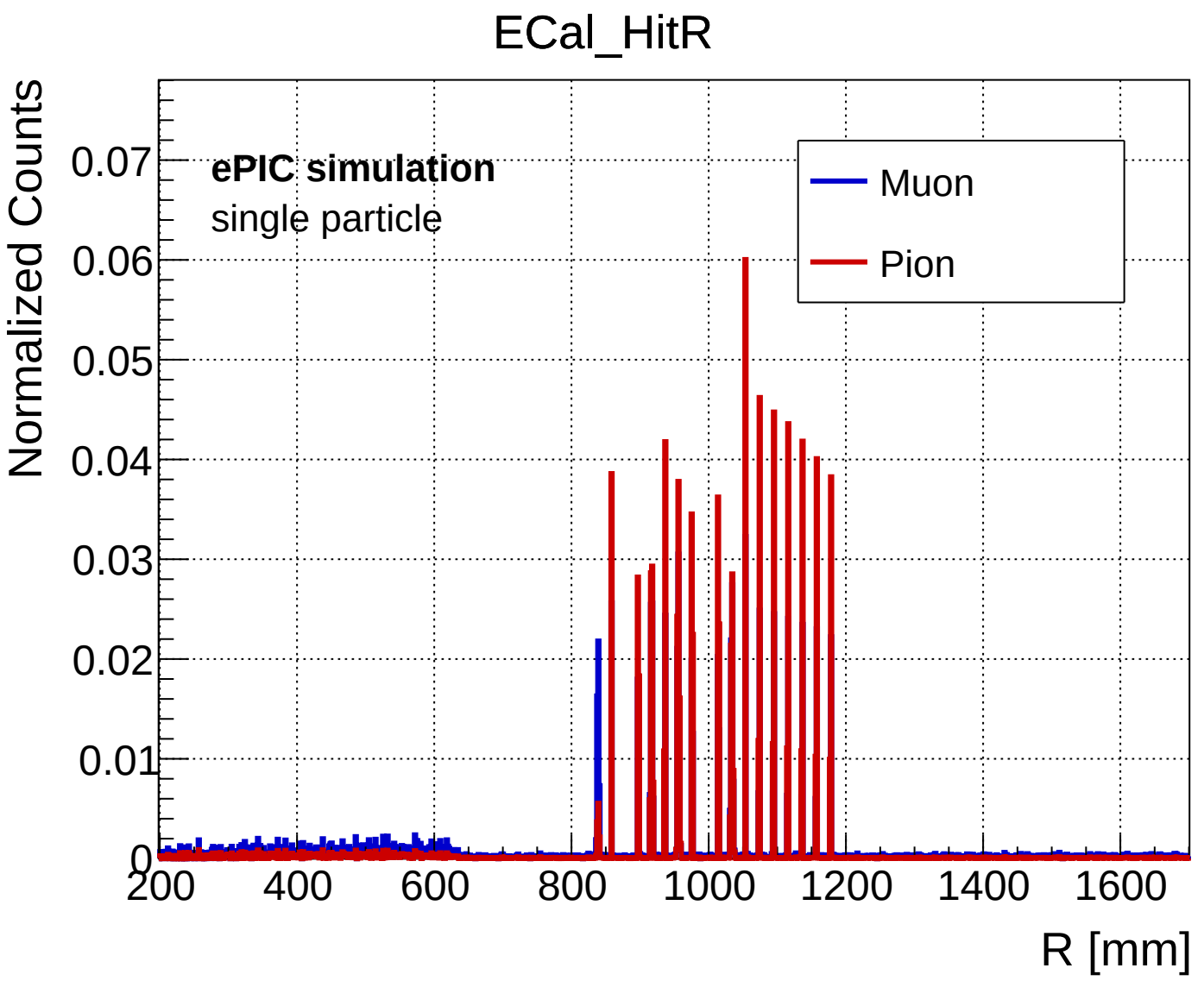
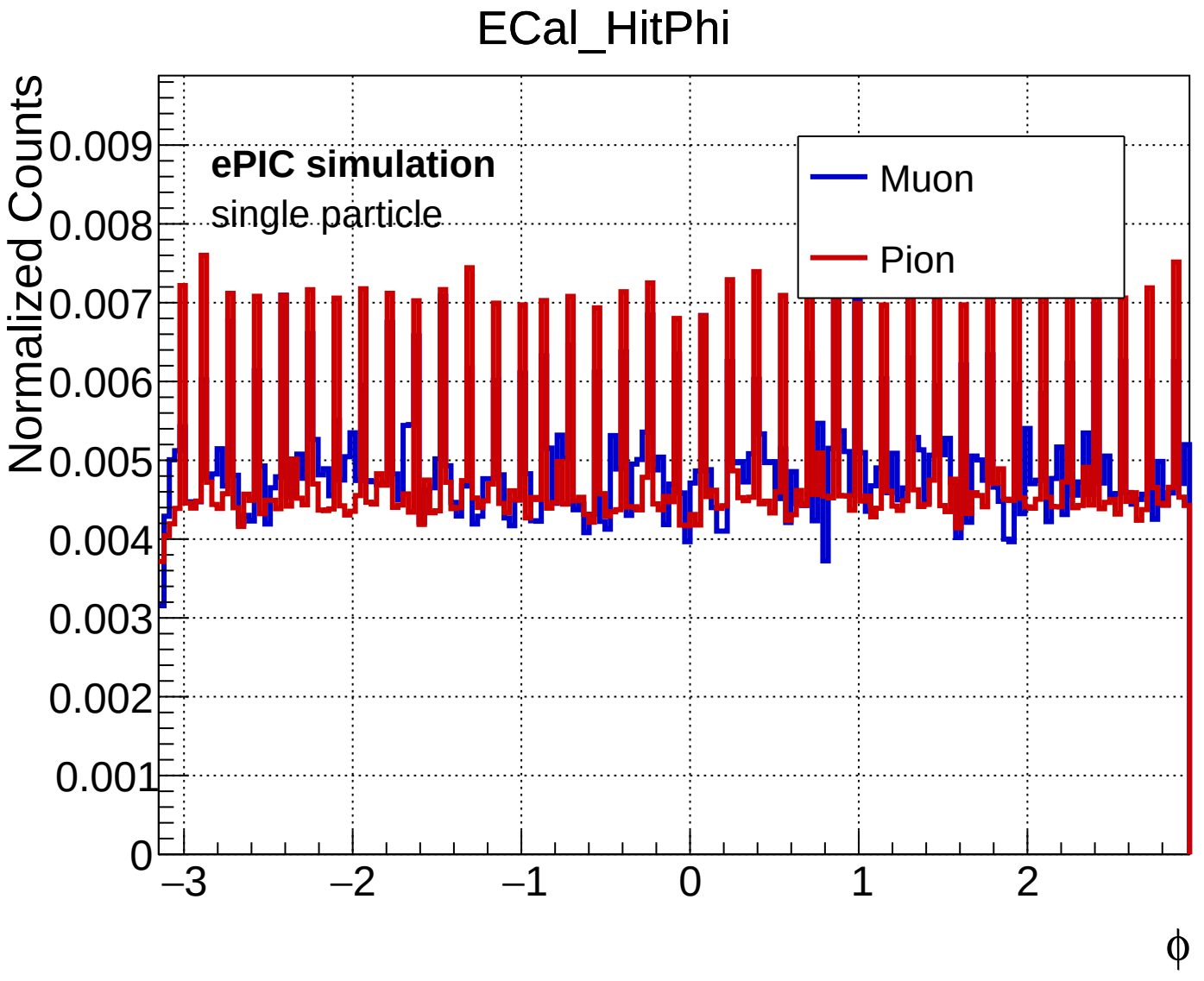
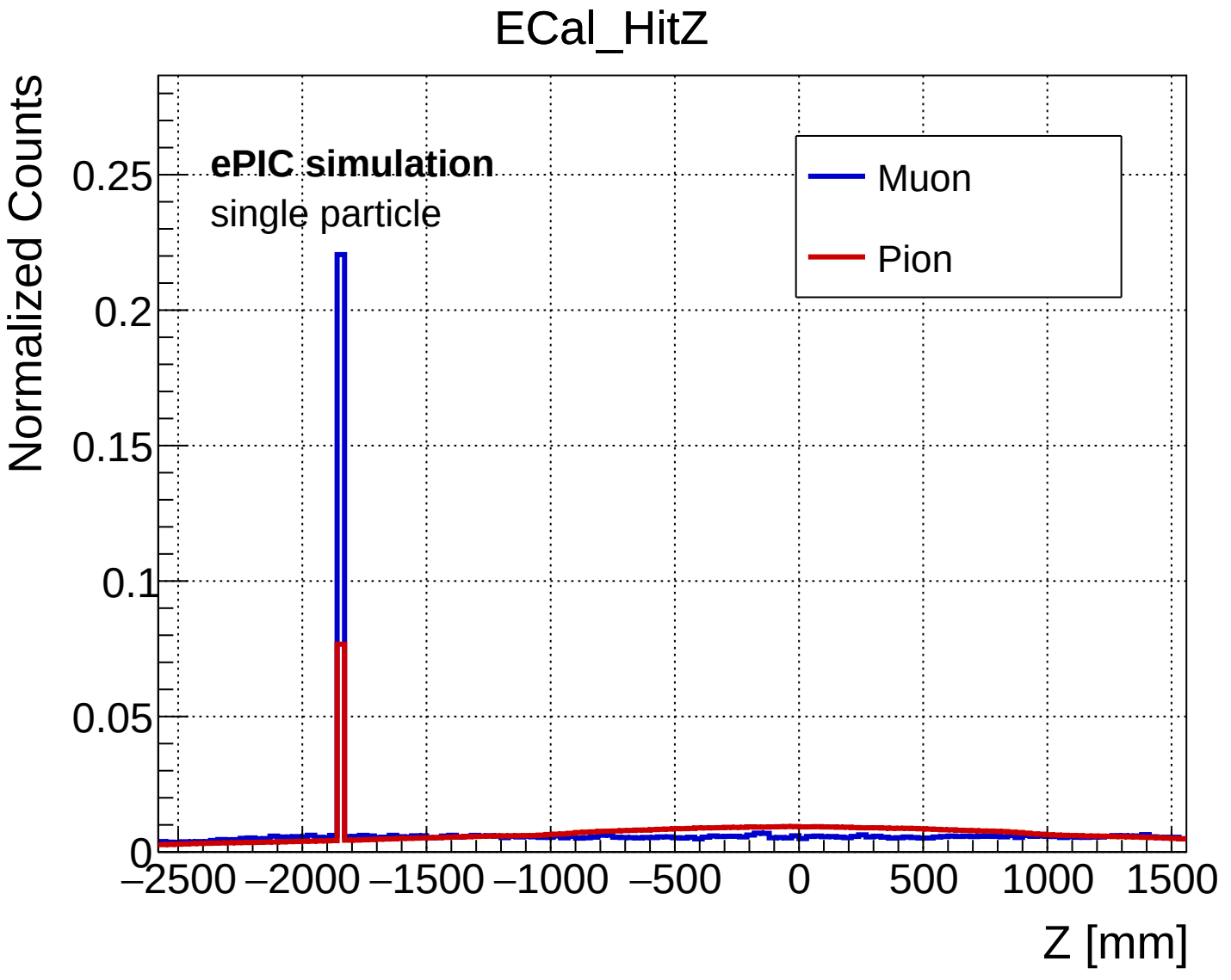
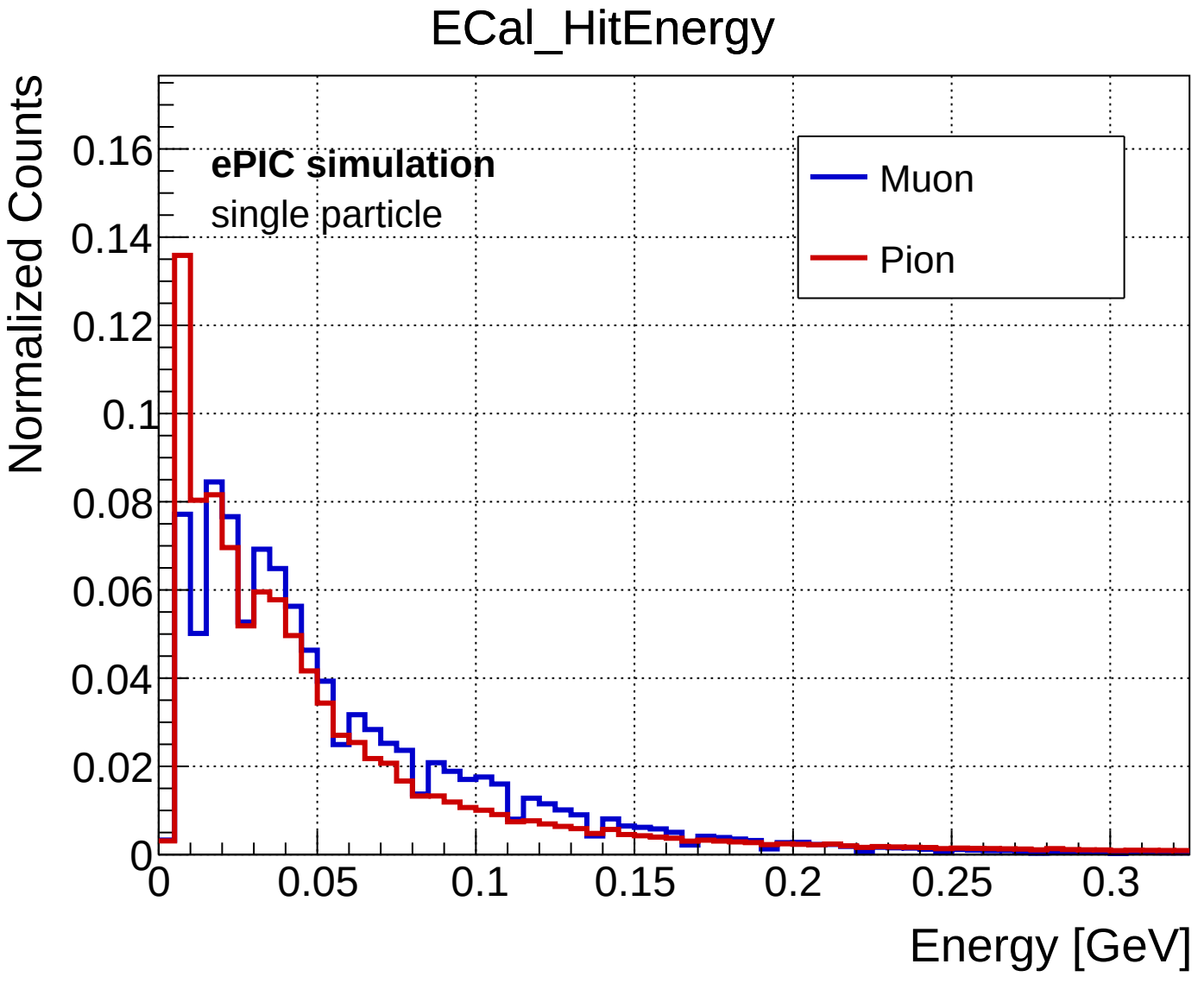
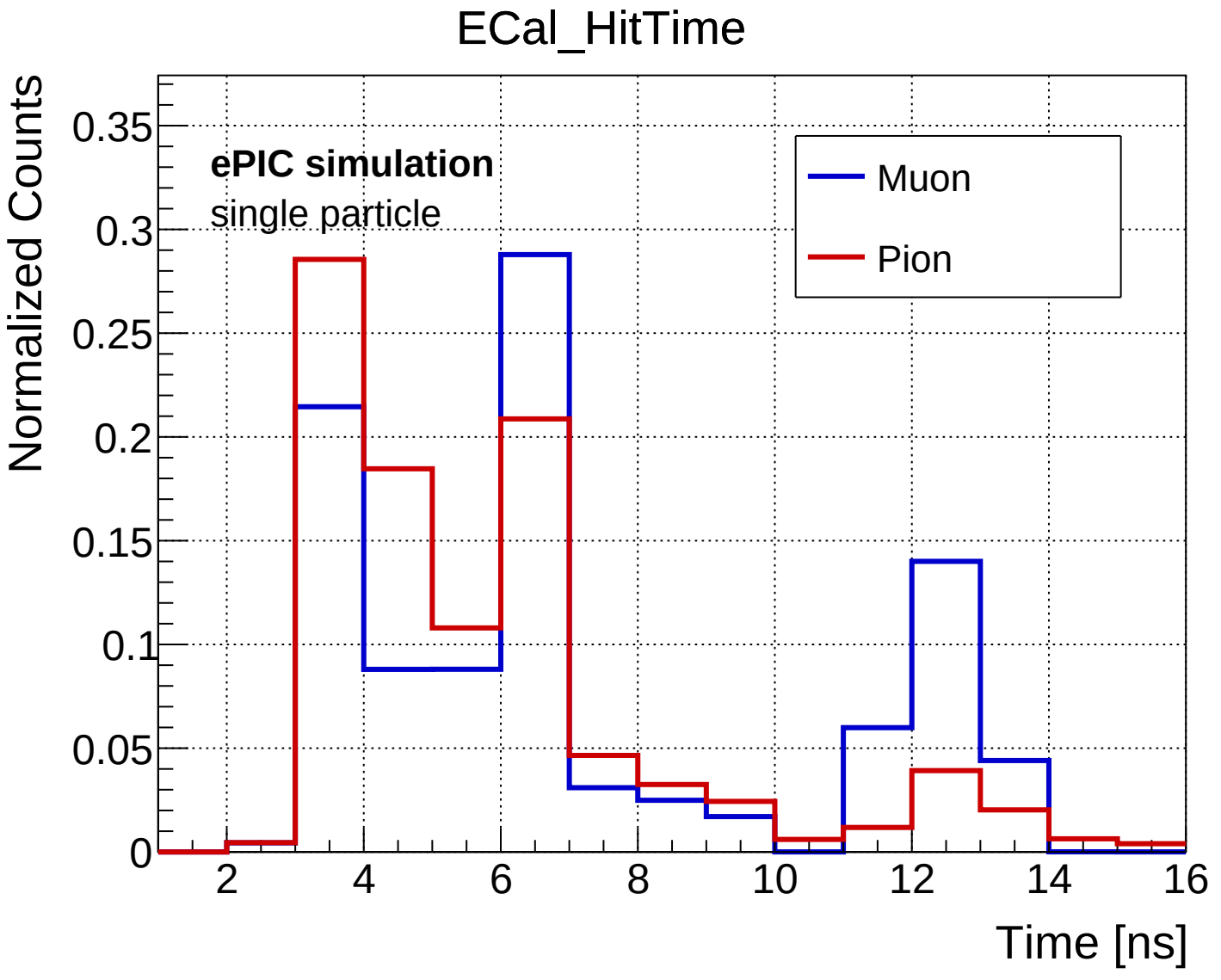
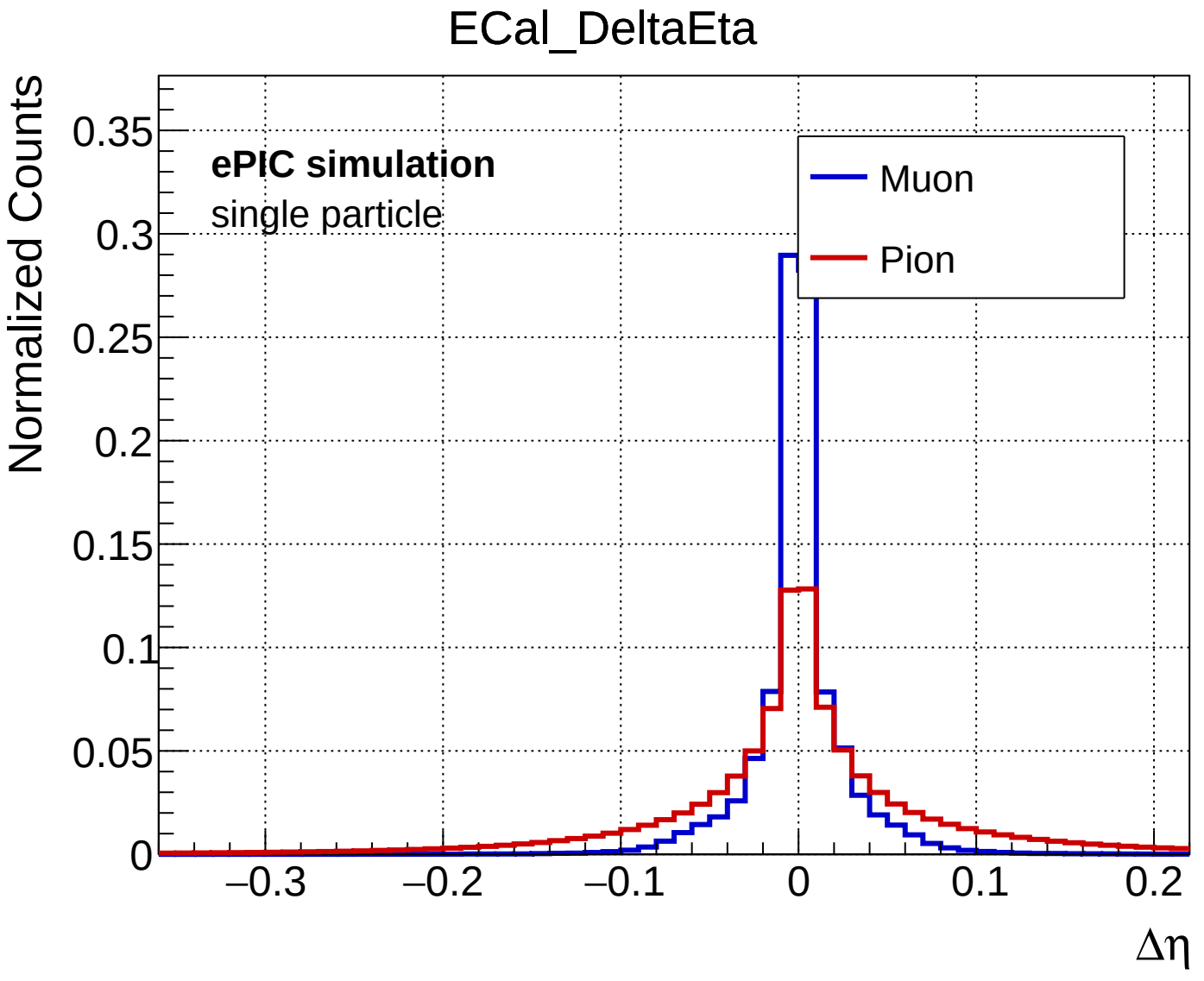

ECal_HitR
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
200
400
600
800
1000
1200
1400
1600
R [mm]

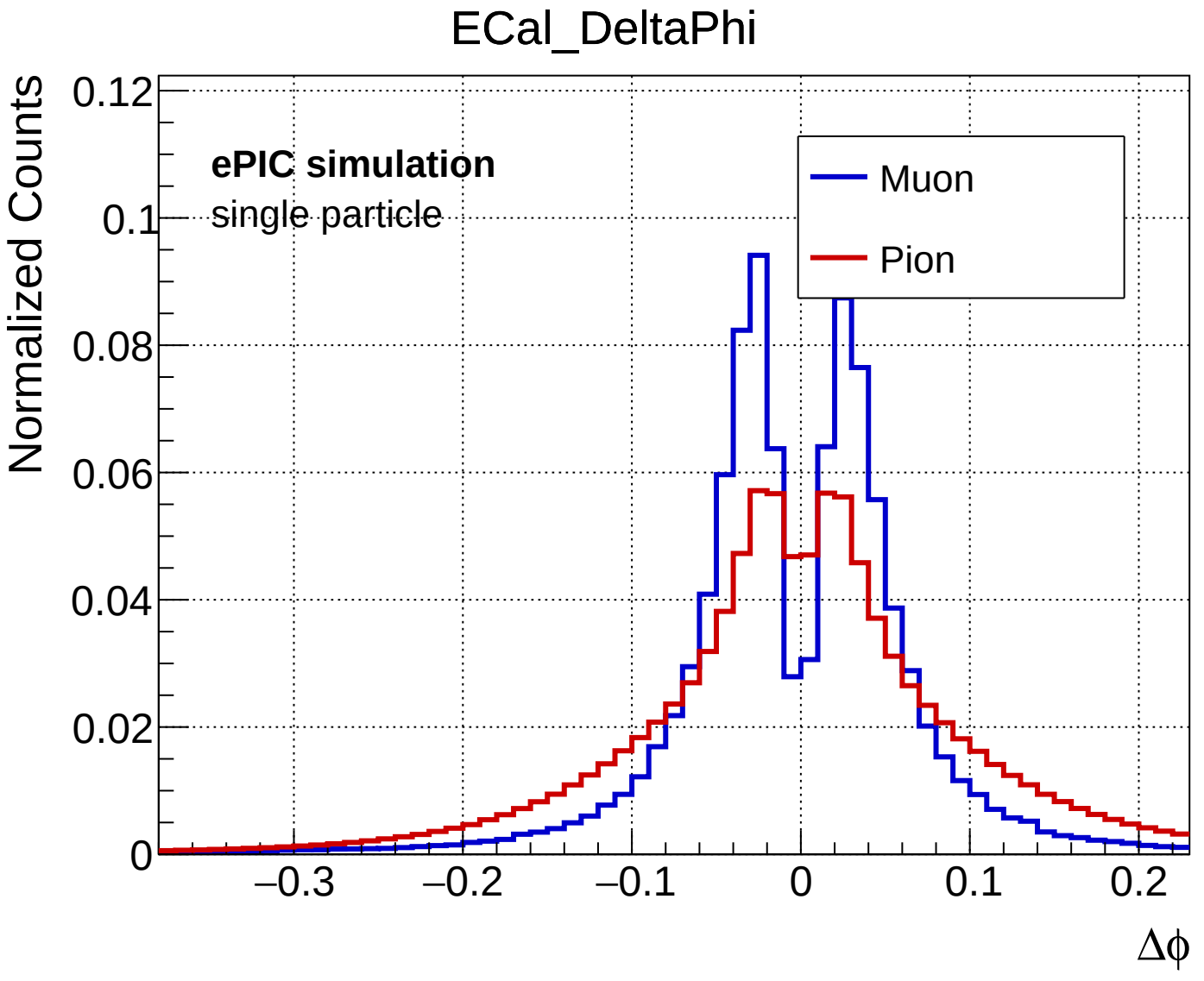

ECal_HitPhi
Normalized Counts
0.009
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
ePIC simulation
single particle
Muon
Pion
−3
−2
−1
0
1
2
ϕ

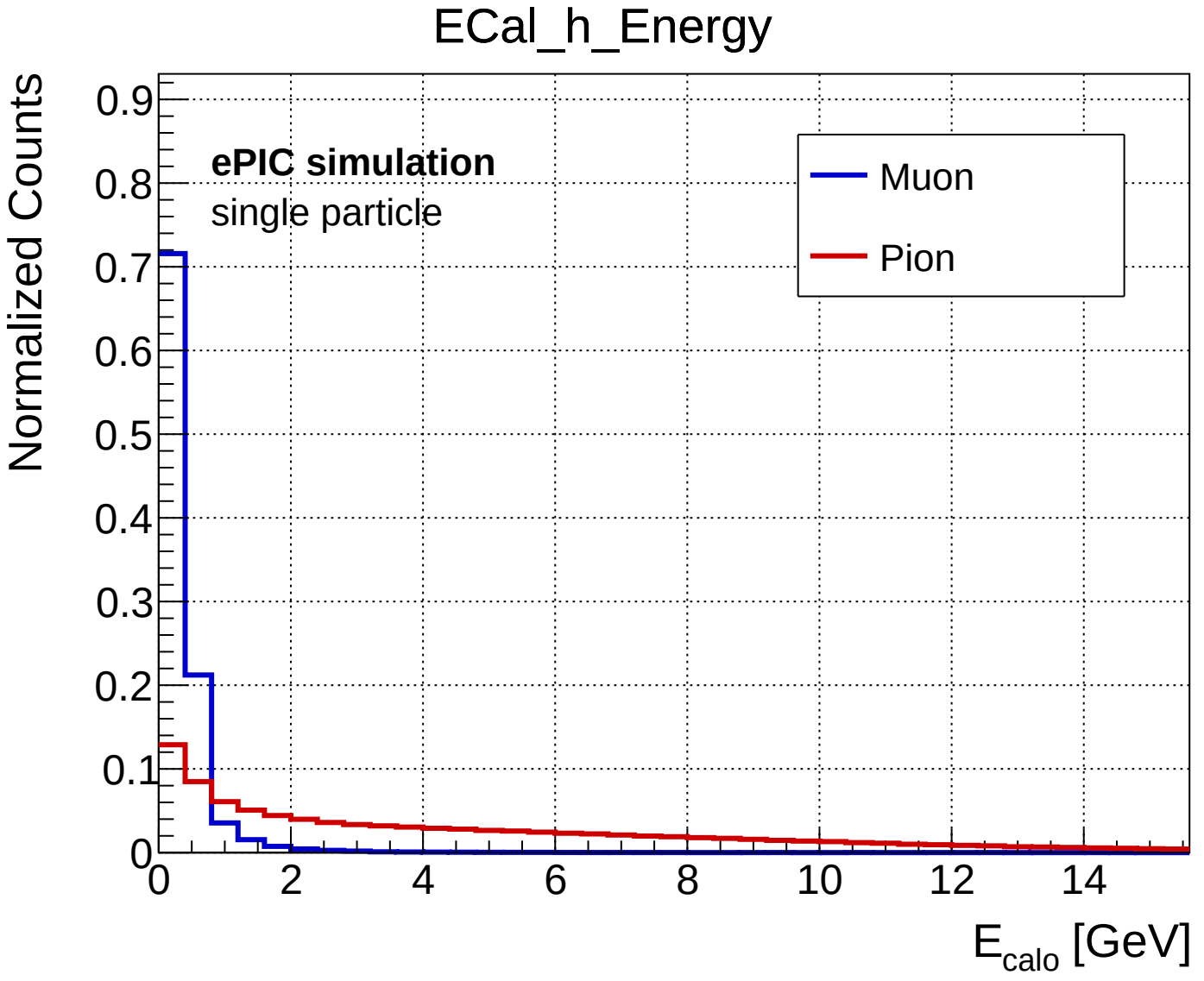

ECal_HitZ
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.25
0.2
0.15
0.1
0.05
0
–2500
–2000
–1500
–1000
–500
0
500
1000
1500
Z [mm]

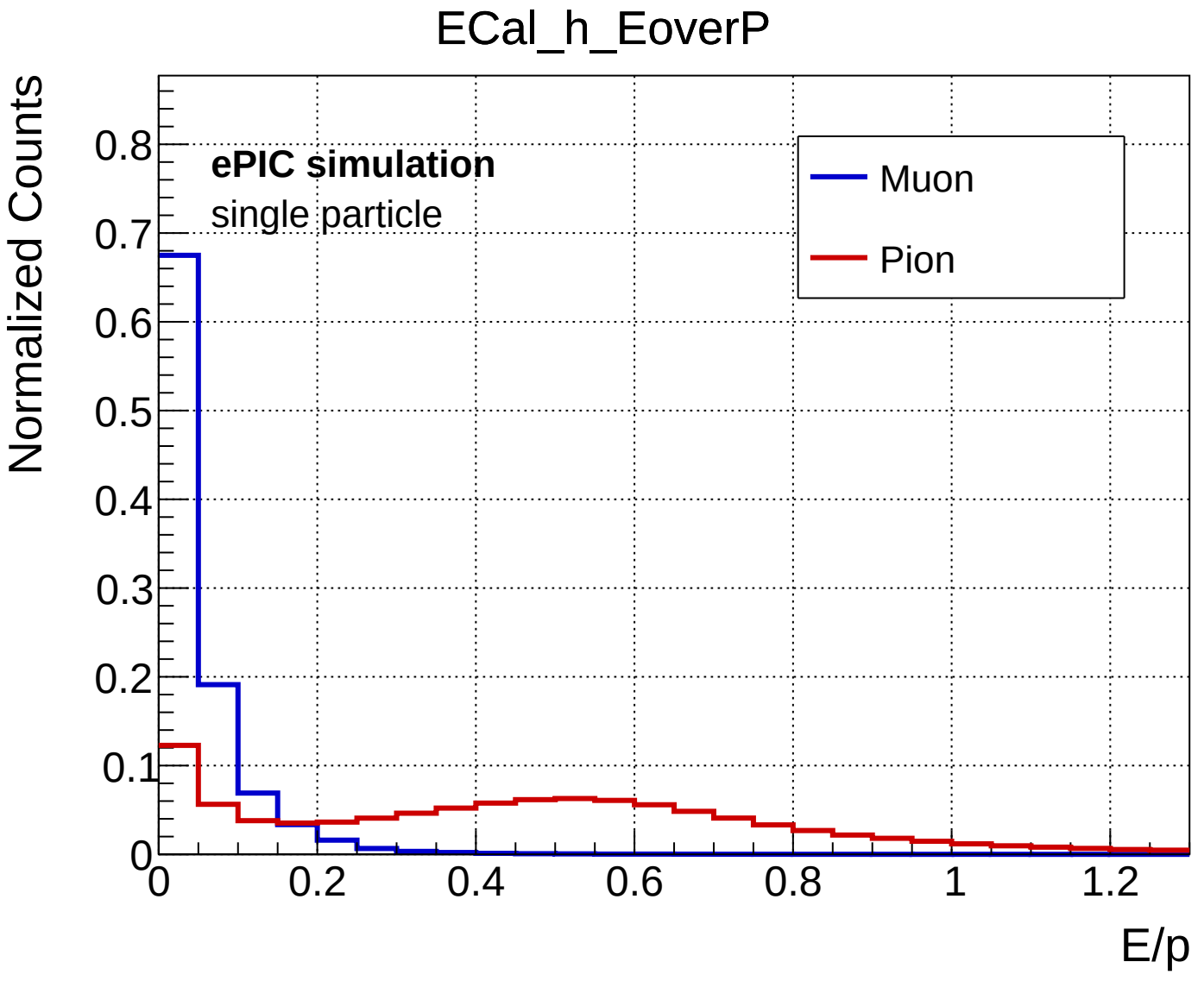

ECal_HitEnergy
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.05
0.1
0.15
0.2
0.25
0.3
Energy [GeV]

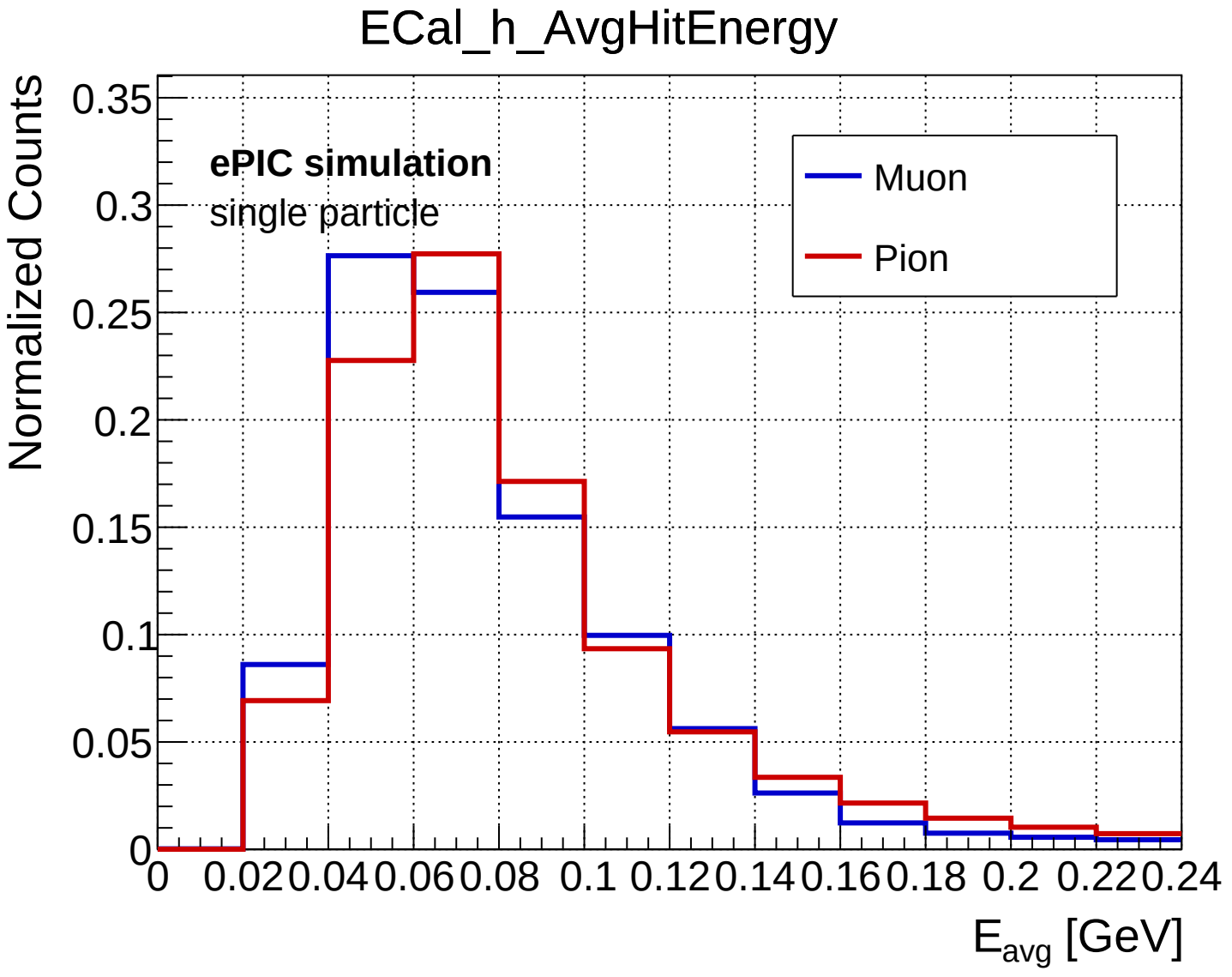

ECal_HitTime
Normalized Counts
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
2
4
6
8
10
12
14
16
Time [ns]

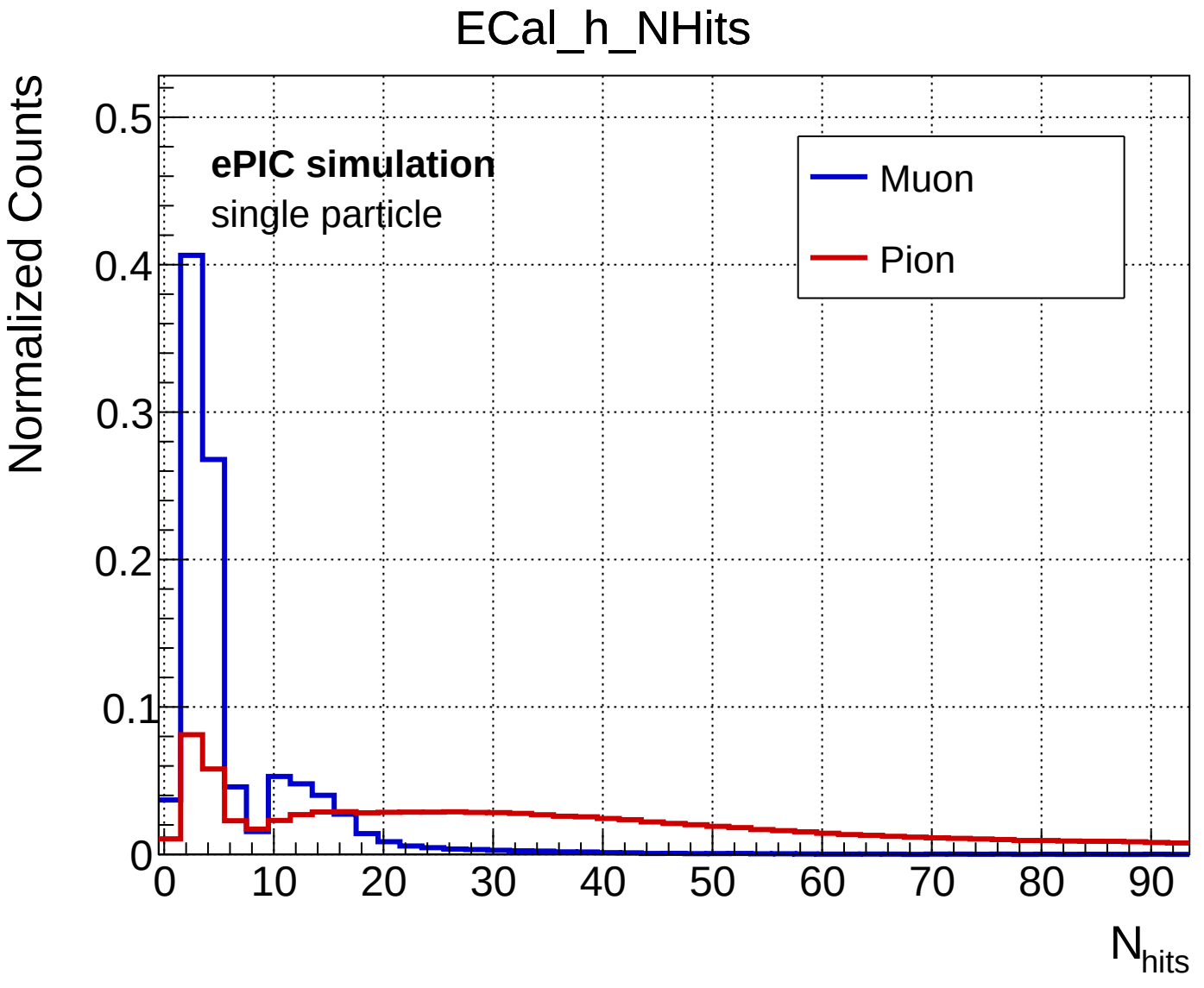

ECal_DeltaEta
Normalized Counts
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
–0.3
–0.2
–0.1
0
0.1
0.2
Δη

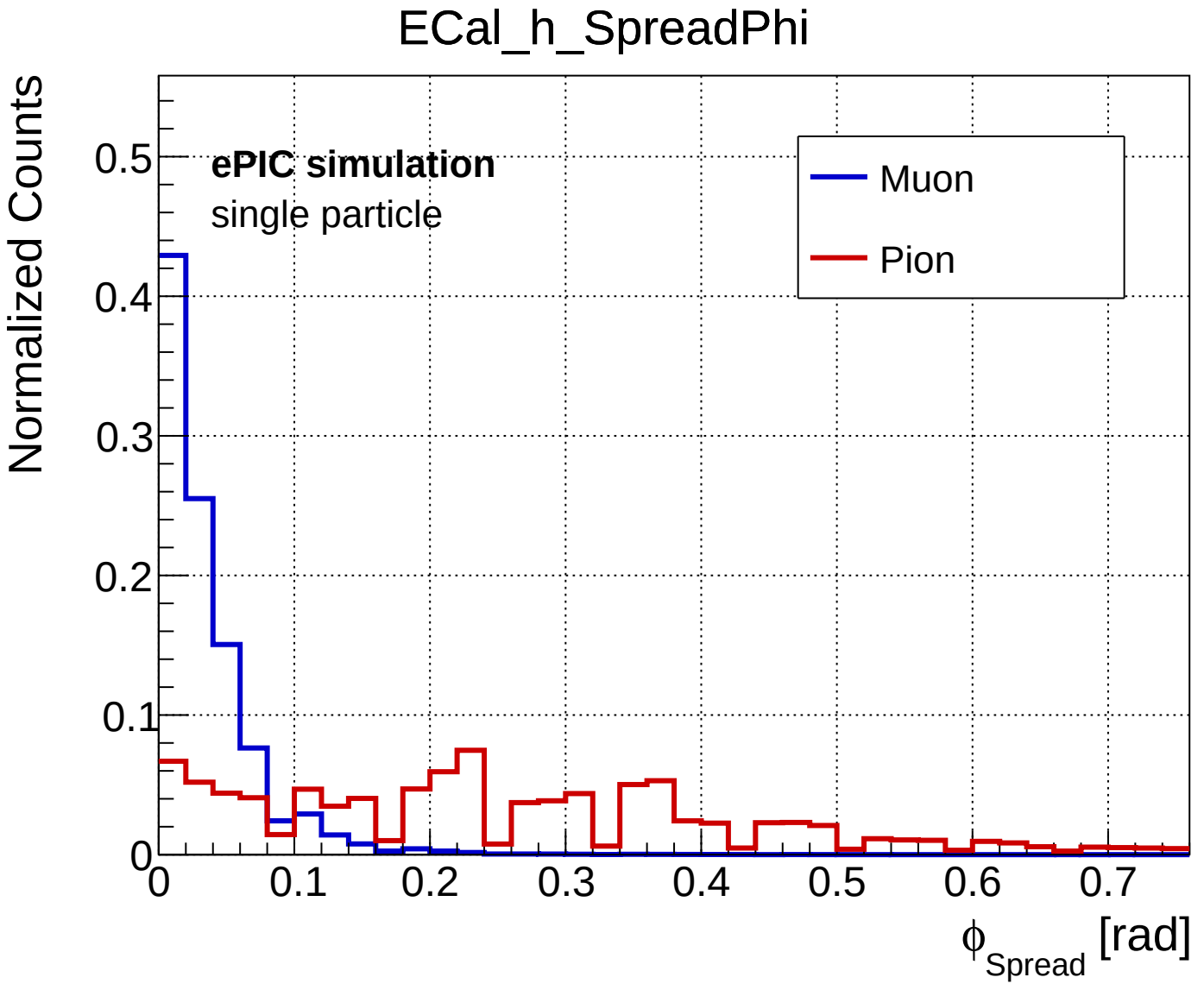

ECal_DeltaPhi
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
–0.3
–0.2
–0.1
0
0.1
0.2
Δϕ

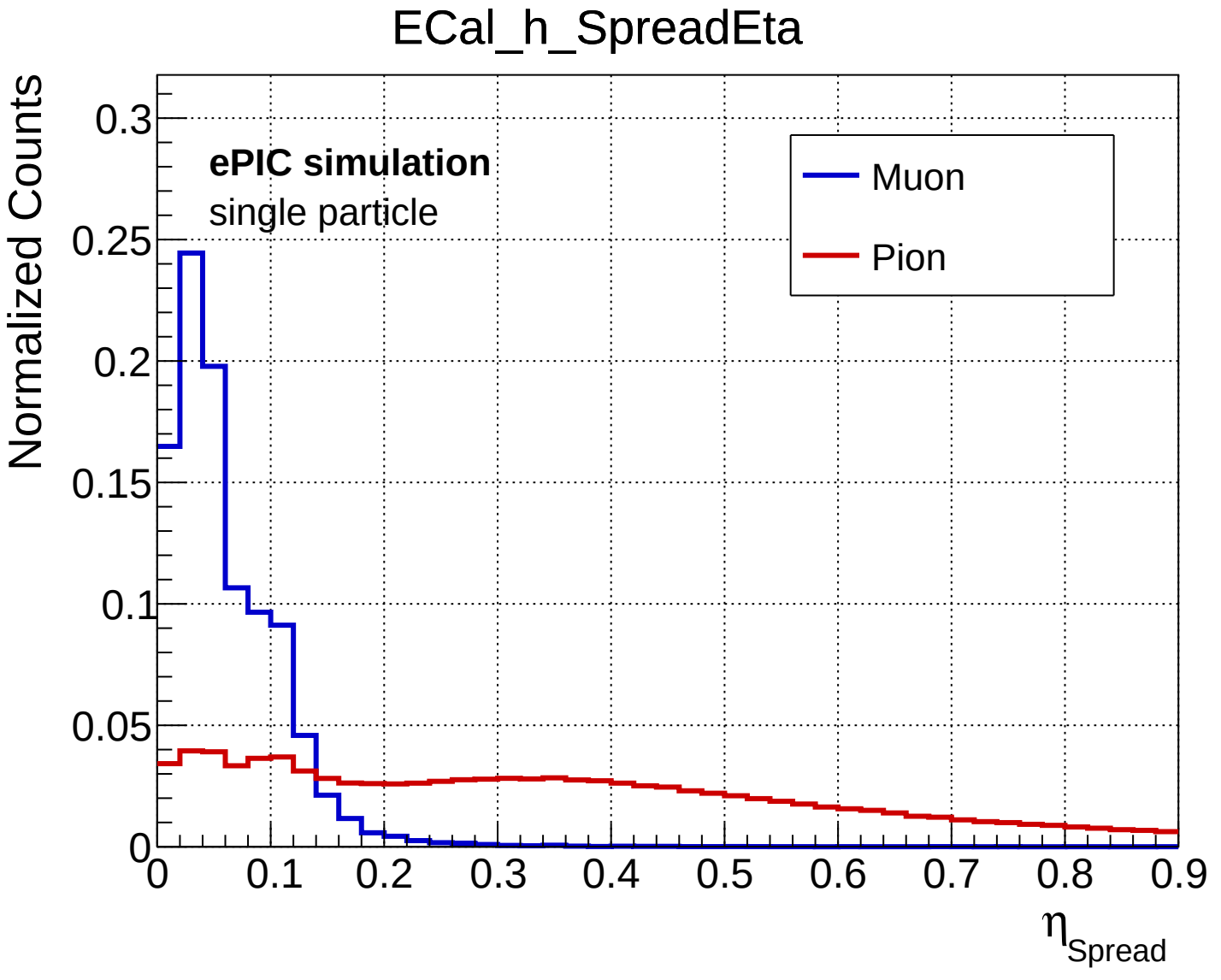

ECal_h_Energy
Normalized Counts
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
2
4
6
8
10
12
14
E_calo [GeV]

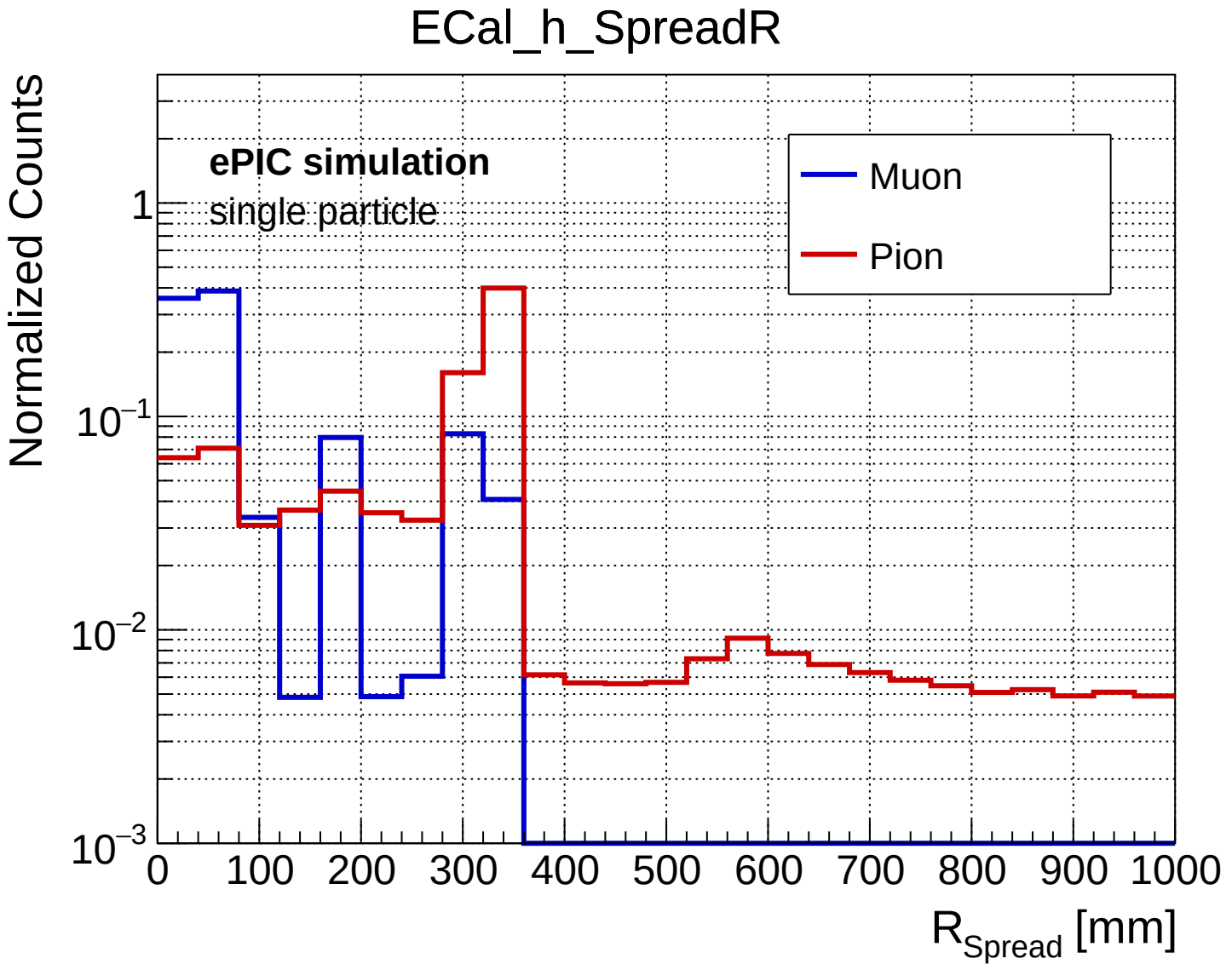

ECal_h_EoverP
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.2
0.4
0.6
0.8
1
1.2
E/p

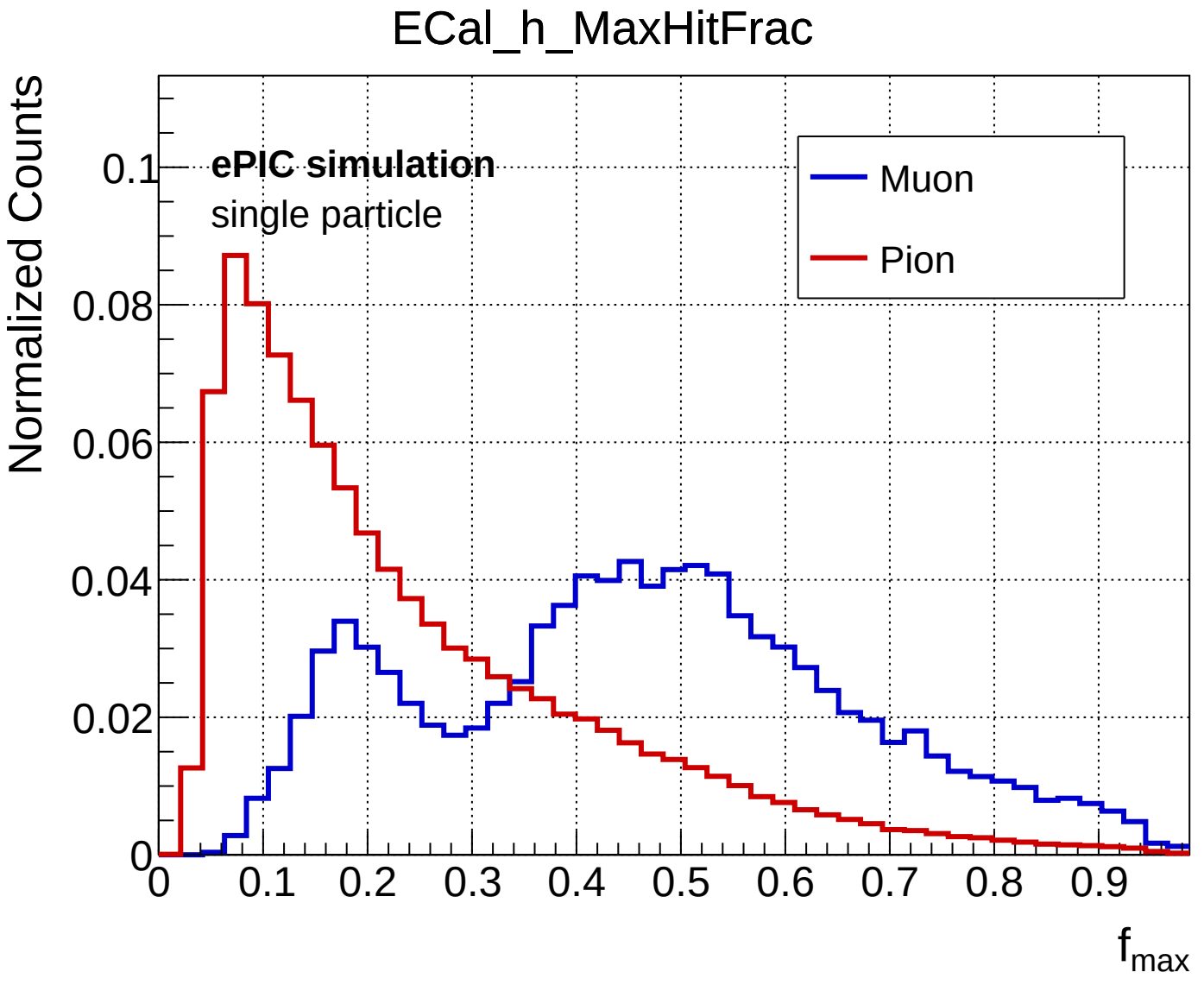

ECal_h_AvgHitEnergy
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
0.22
0.24
E_avg [GeV]

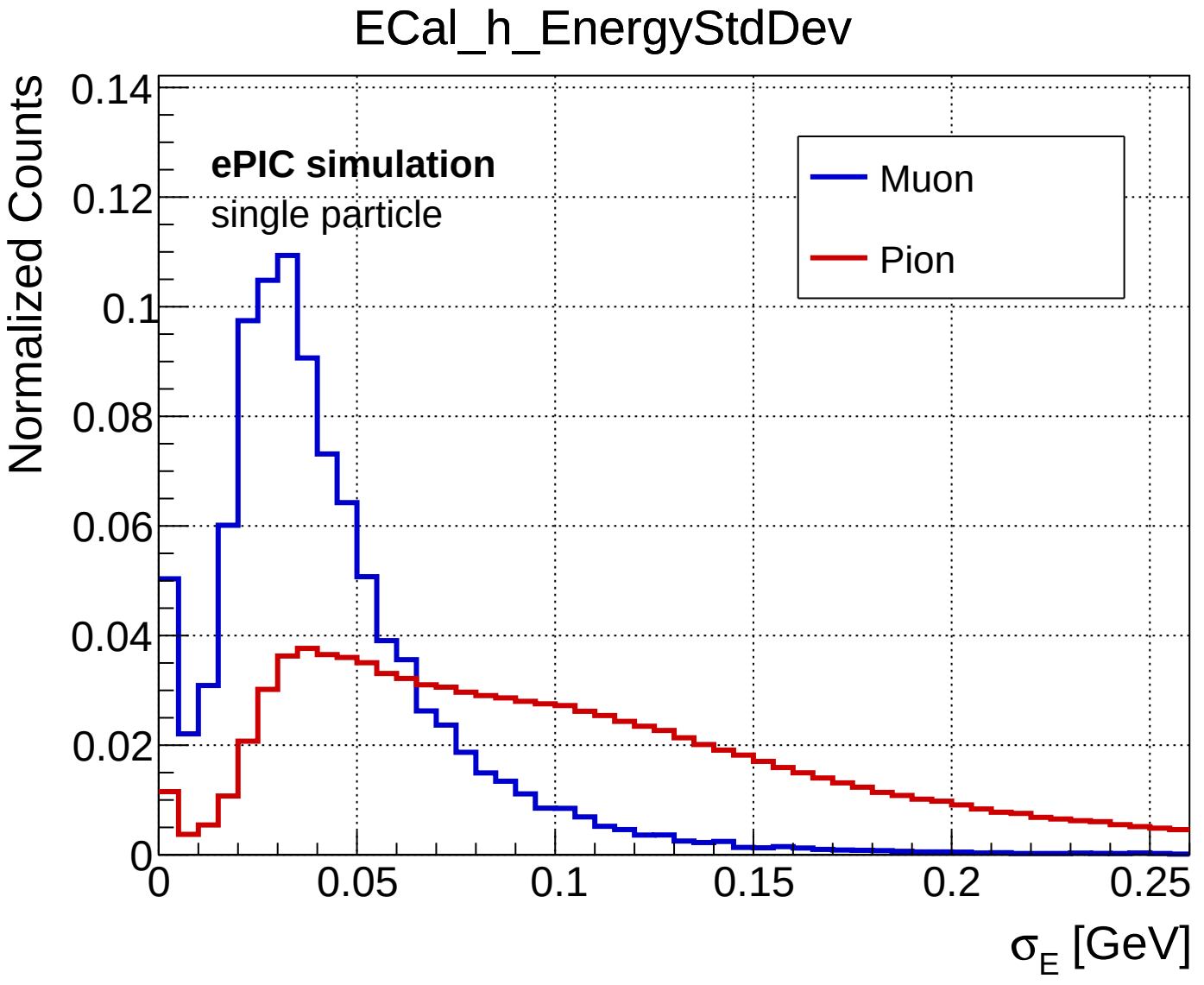

ECal_h_NHits
Normalized Counts
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
10
20
30
40
50
60
70
80
90
N_hits

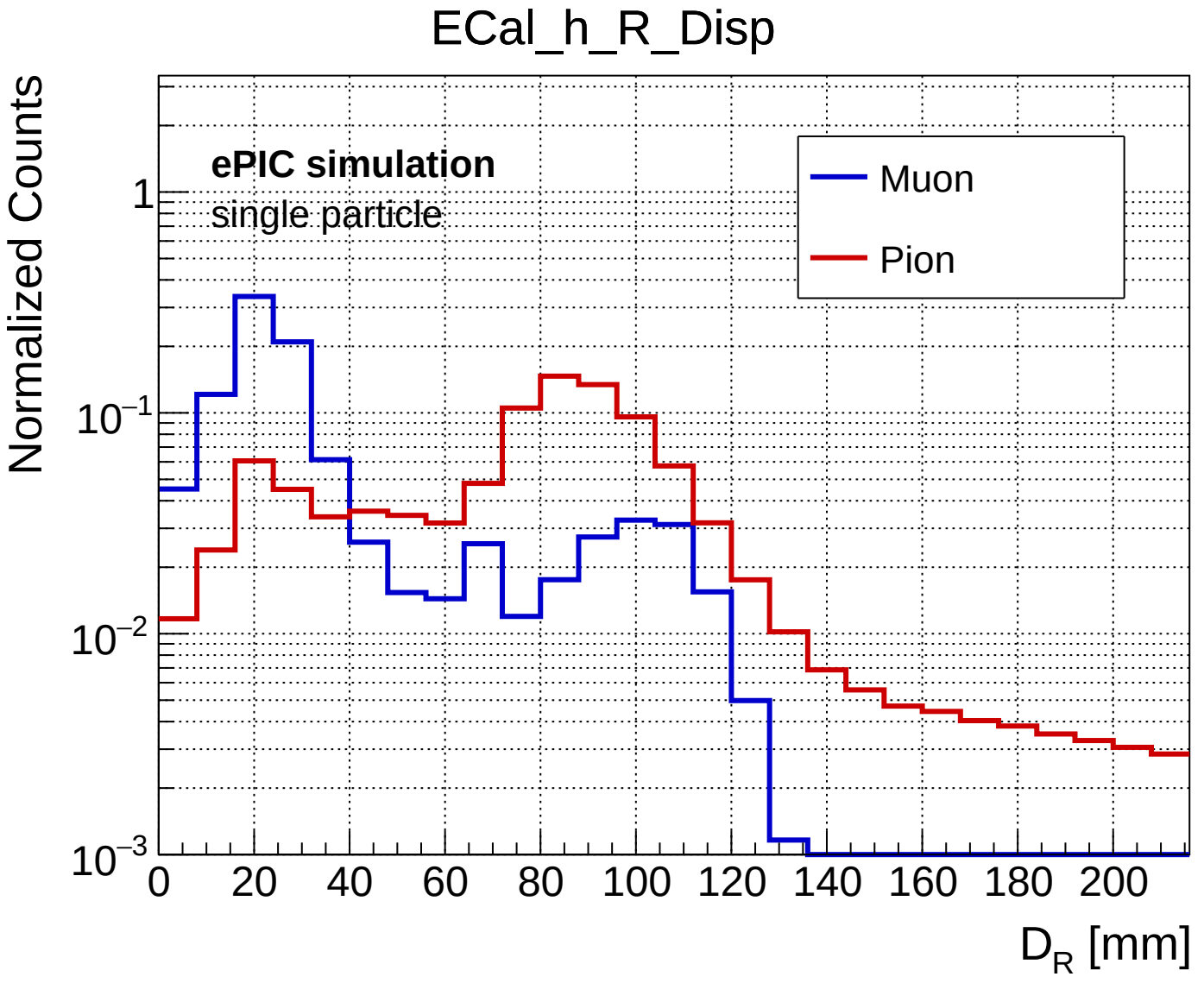

ECal_h_SpreadPhi
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
ϕ_{Spread} [rad]

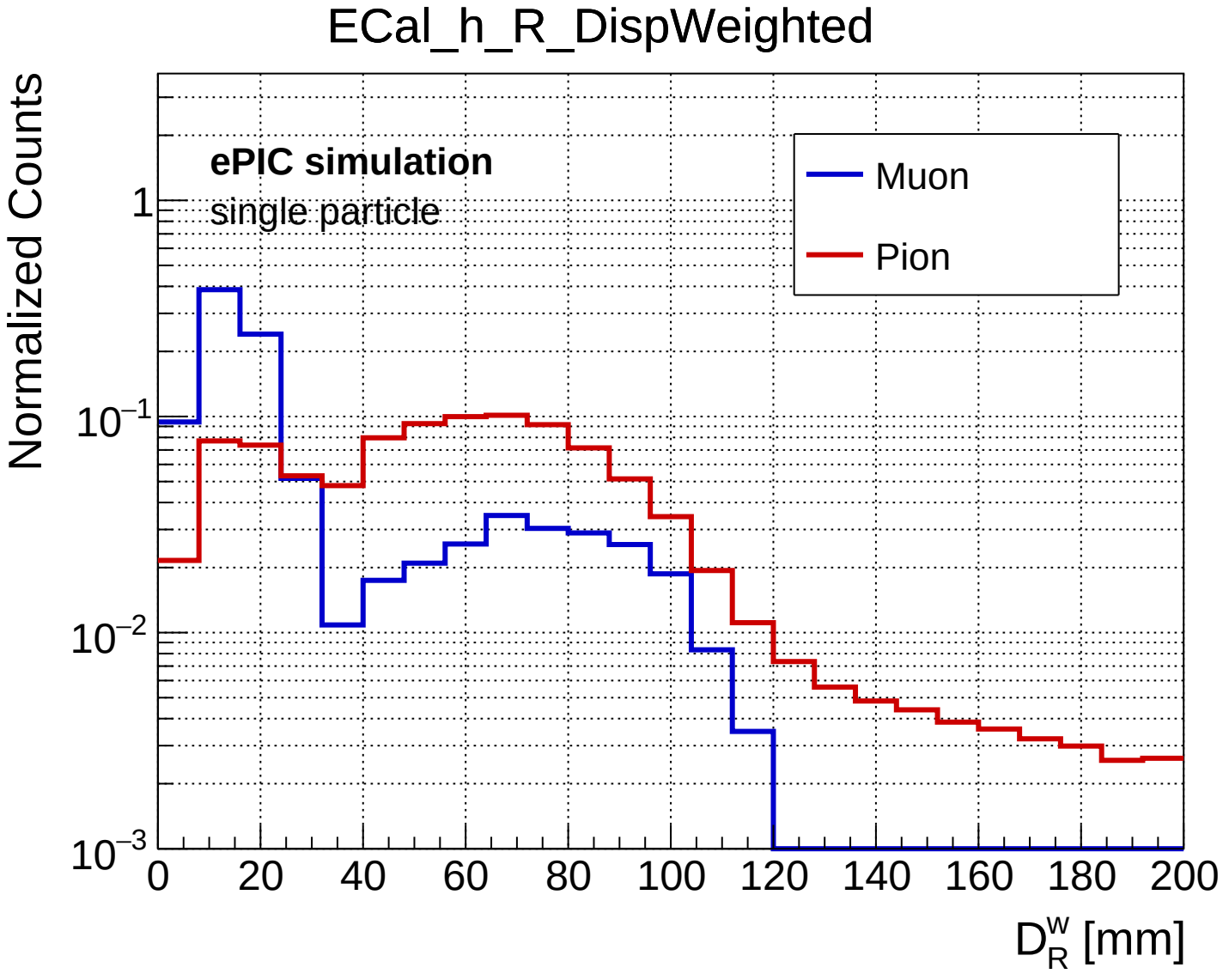

ECal_h_SpreadEta
Normalized Counts
0.3
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
η Spread

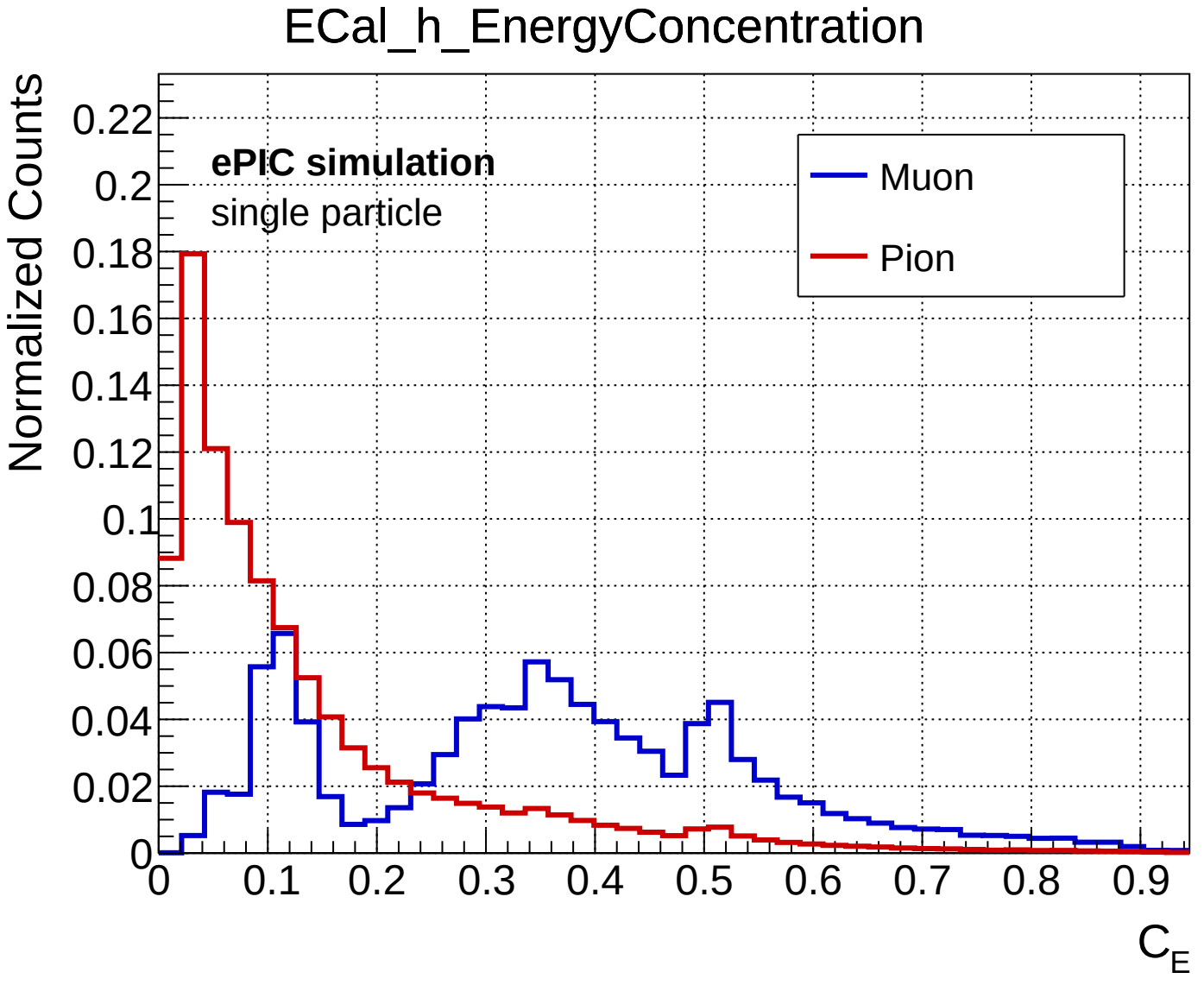

ECal_h_SpreadR
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
100
200
300
400
500
600
700
800
900
1000
R_{Spread} [mm]

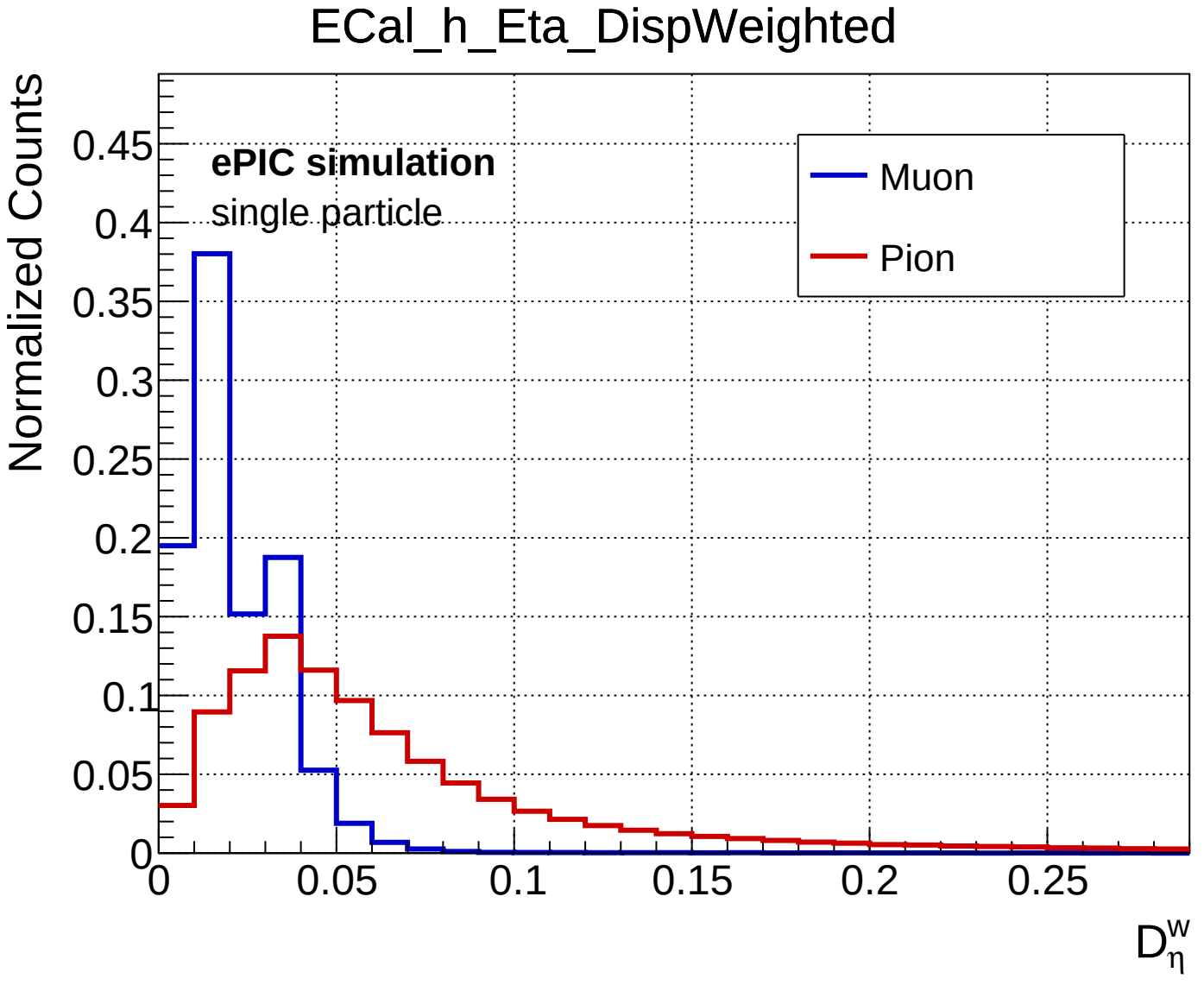

ECal_h_MaxHitFrac
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.1
0.08
0.06
0.04
0.02
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
f_max

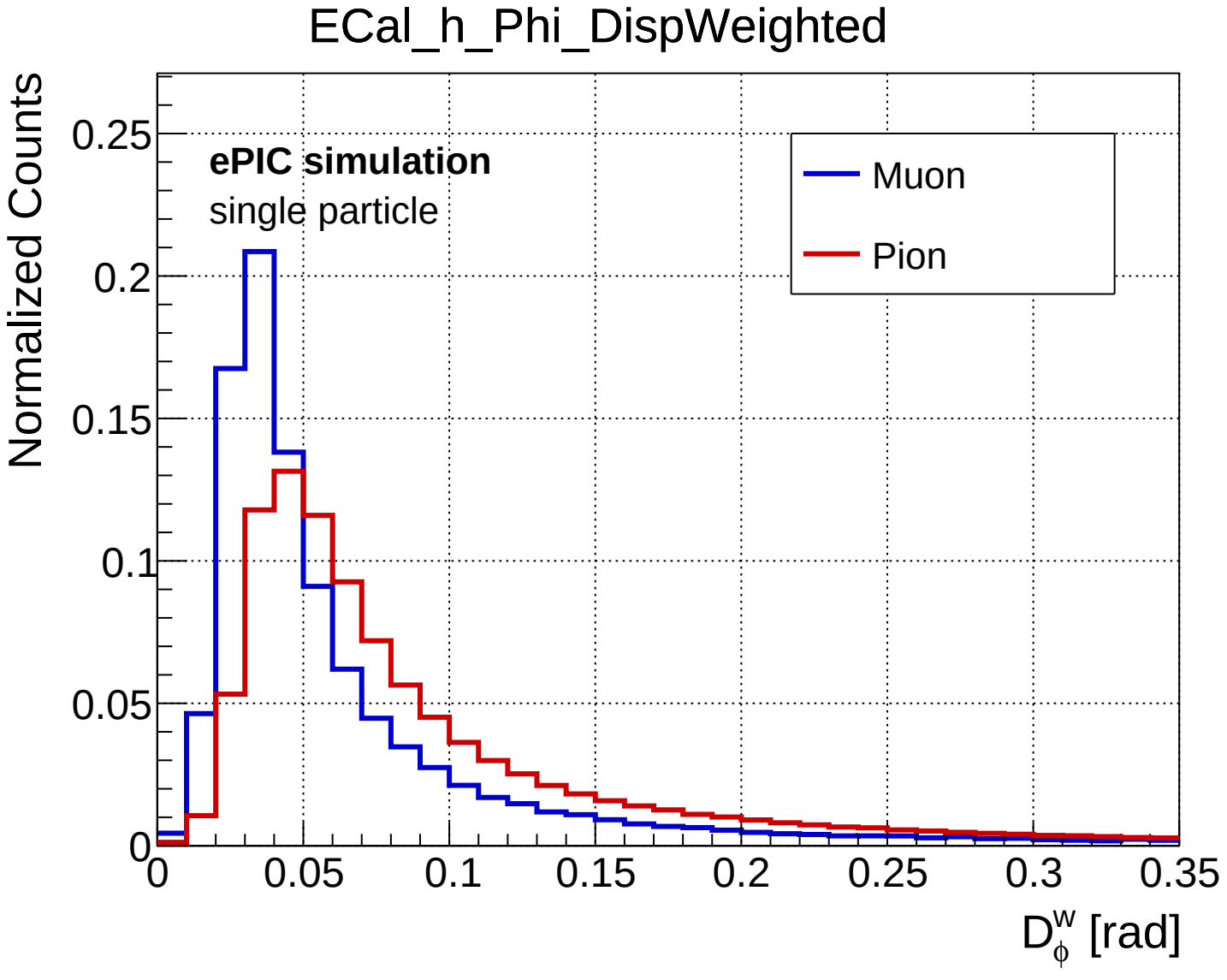

ECal_h_EnergyStdDev
Normalized Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.05
0.1
0.15
0.2
0.25
σE [GeV]
ePIC simulation
single particle
Muon
Pion

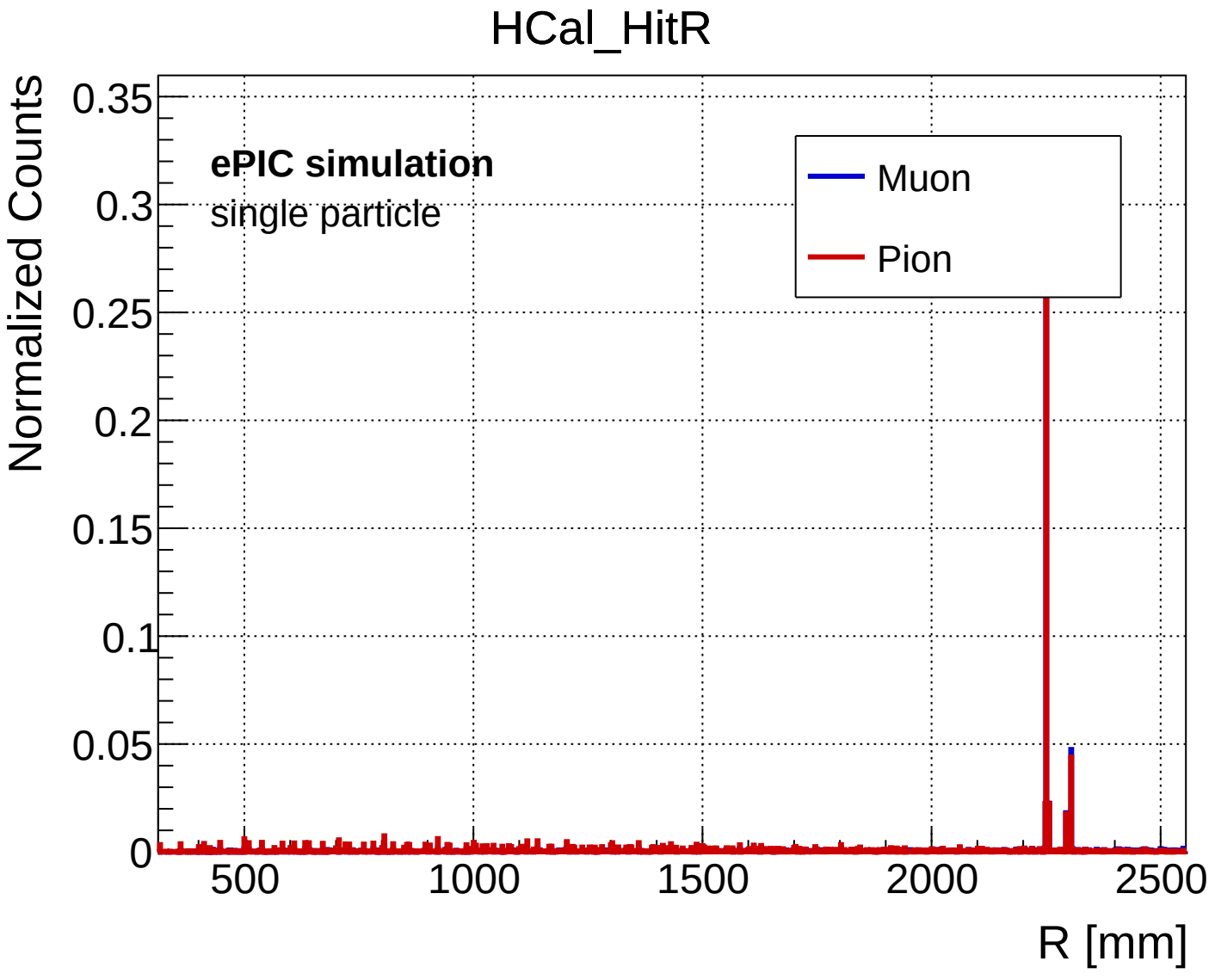

ECal_h_R_Disp
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
120
140
160
180
200
D_R [mm]

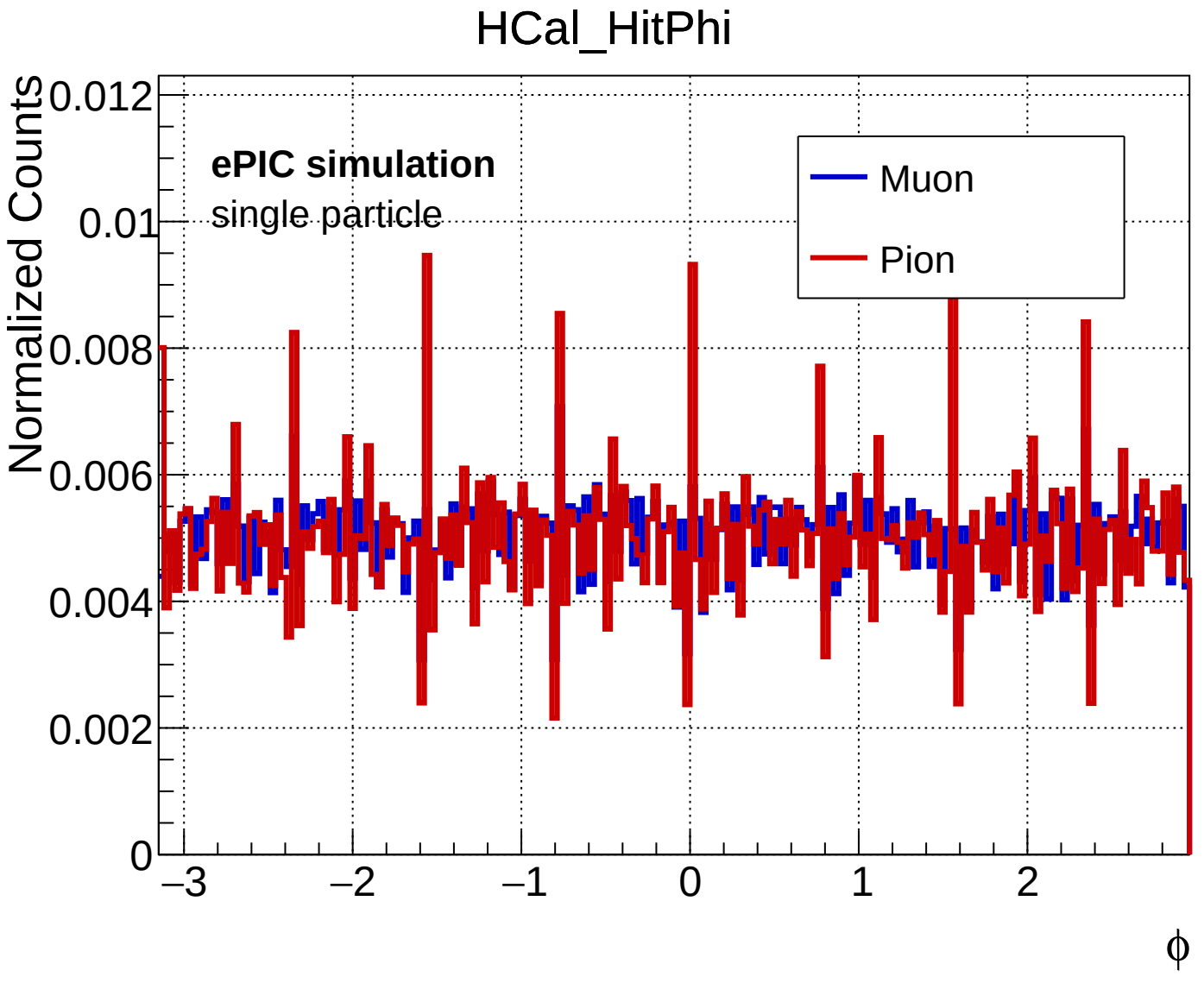

ECal_h_R_DispWeighted
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
20
40
60
80
100
120
140
160
180
200
D_R^w [mm]

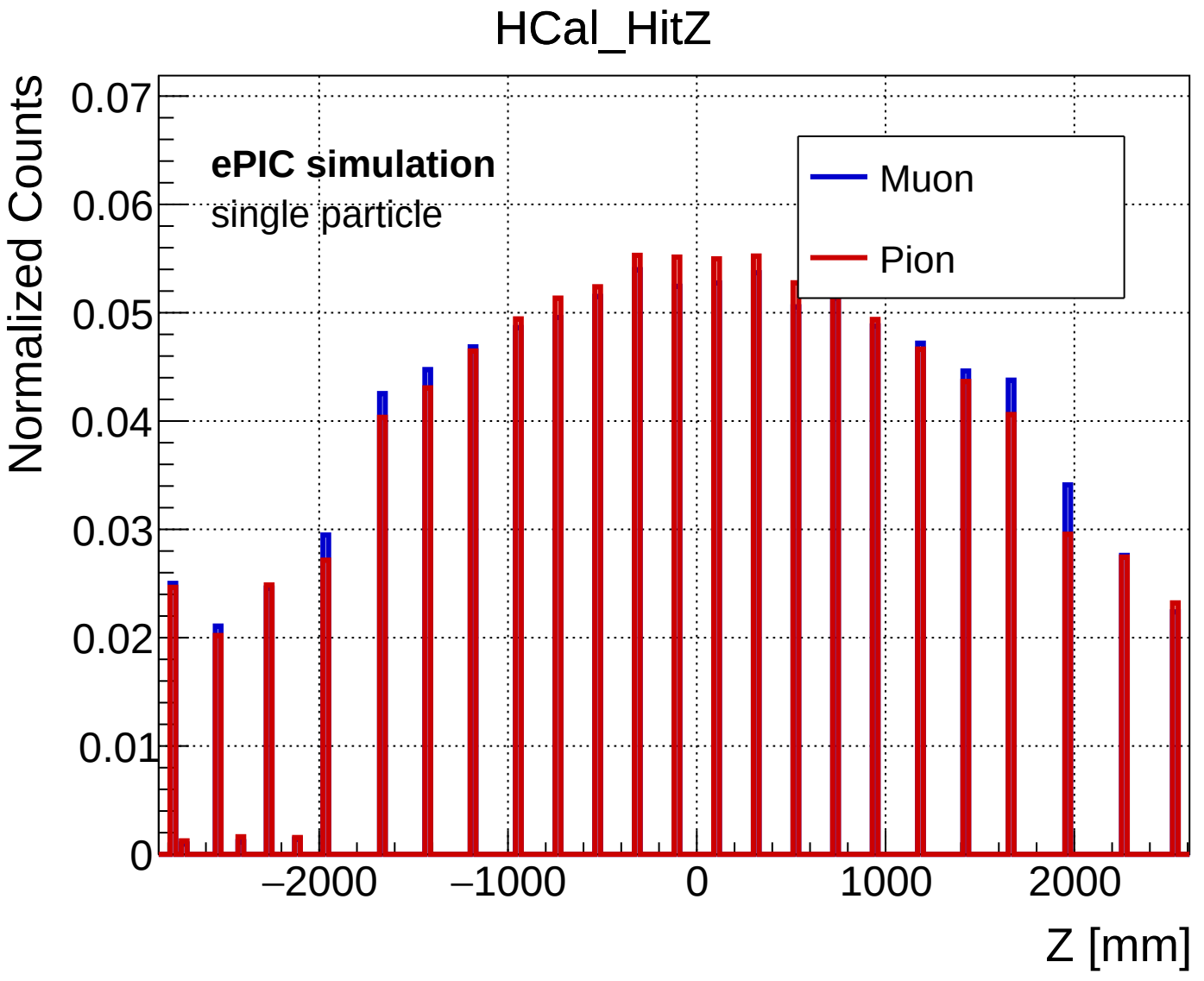

ECal_h_EnergyConcentration
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
C_E

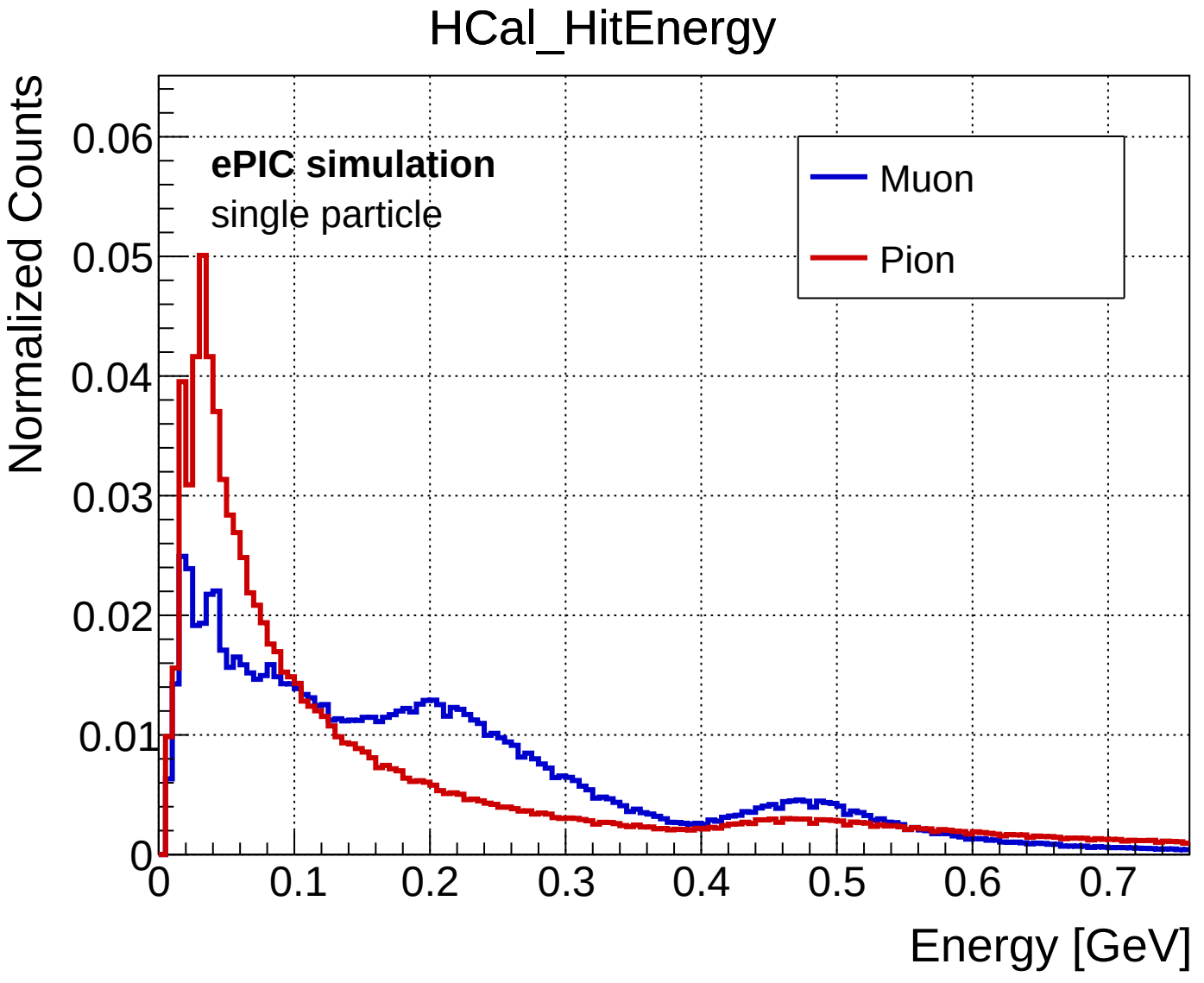

ECal_h_Eta_DispWeighted
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.05
0.1
0.15
0.2
0.25
D_η^w
ePIC simulation
single particle
Muon
Pion

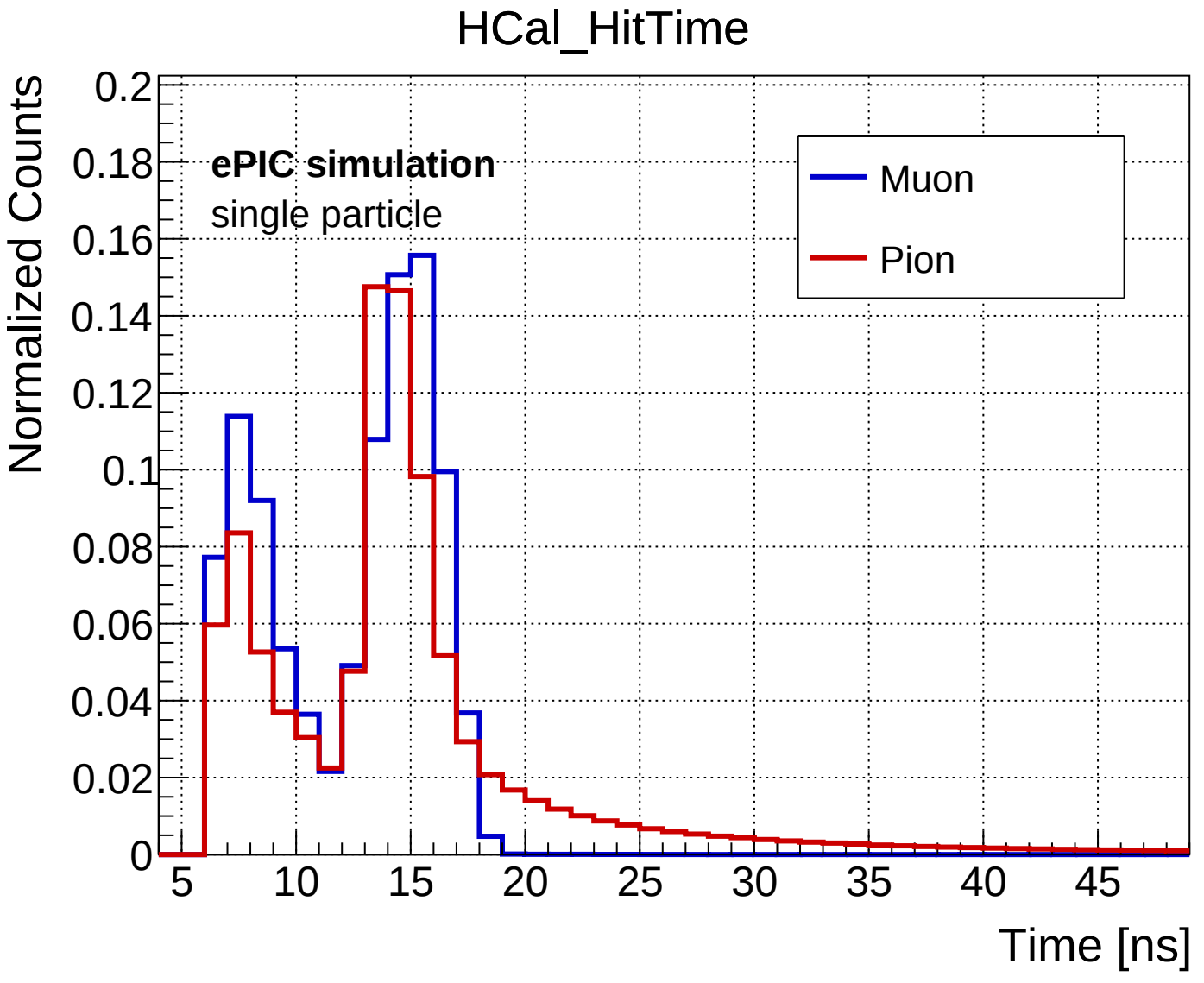

ECal_h_Phi_DispWeighted
Normalized Counts
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
D_{ϕ}^{w} [rad]

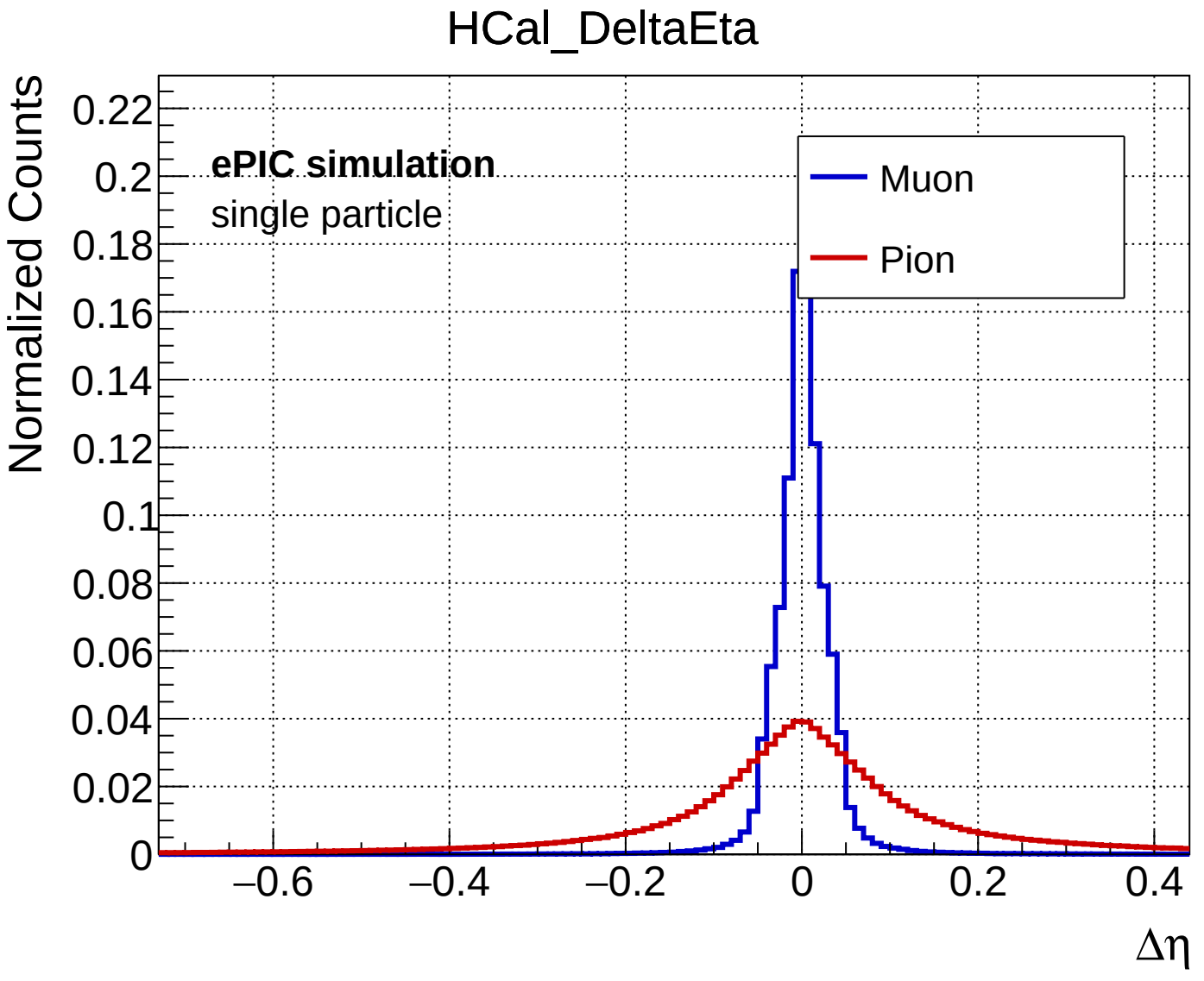

HCal_HitR
Normalized Counts
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
500
1000
1500
2000
2500
R [mm]

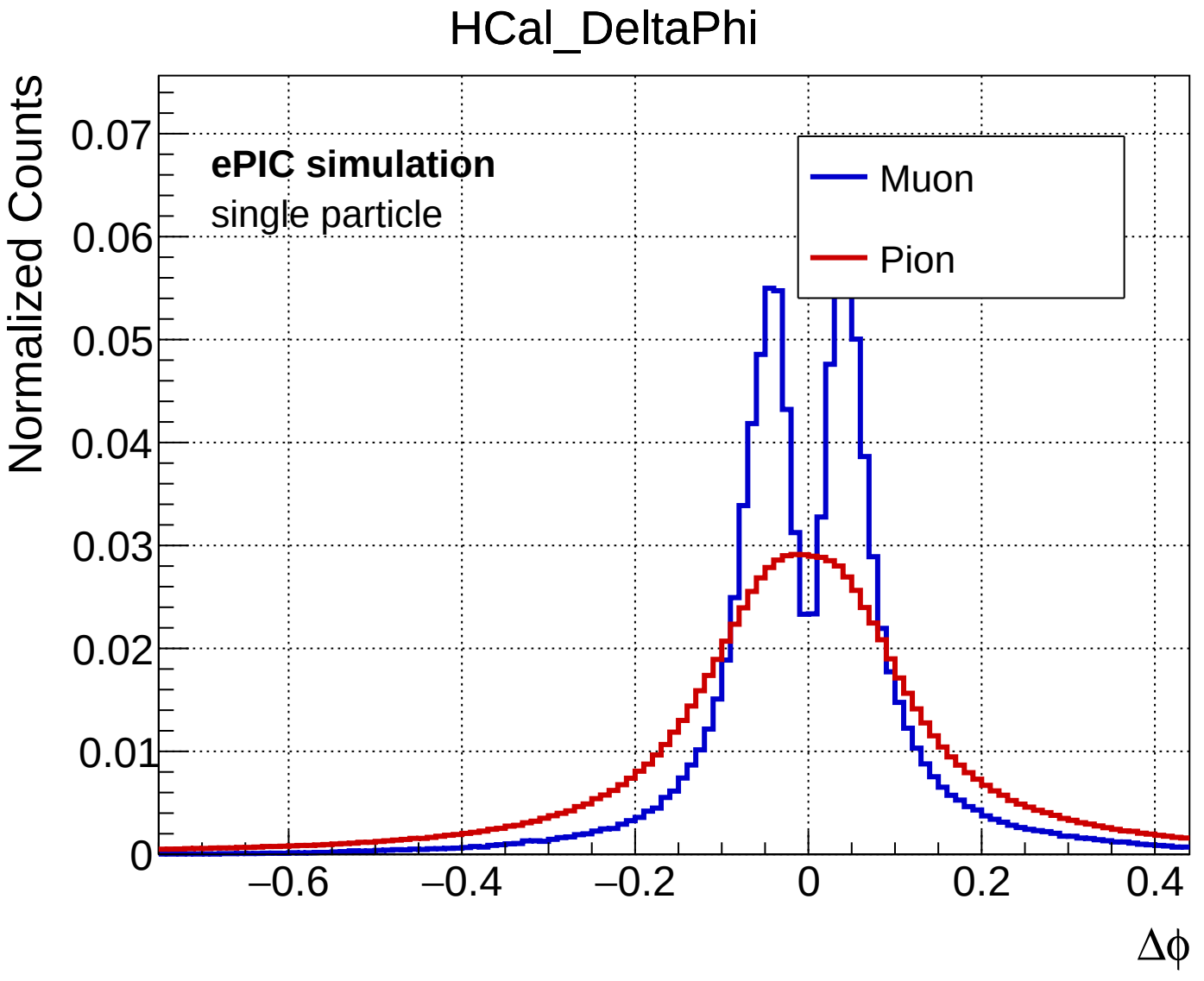

HCal_HitPhi
Normalized Counts
0.012
0.01
0.008
0.006
0.004
0.002
0
−3
−2
−1
0
1
2
ϕ
ePIC simulation
single particle
Muon
Pion

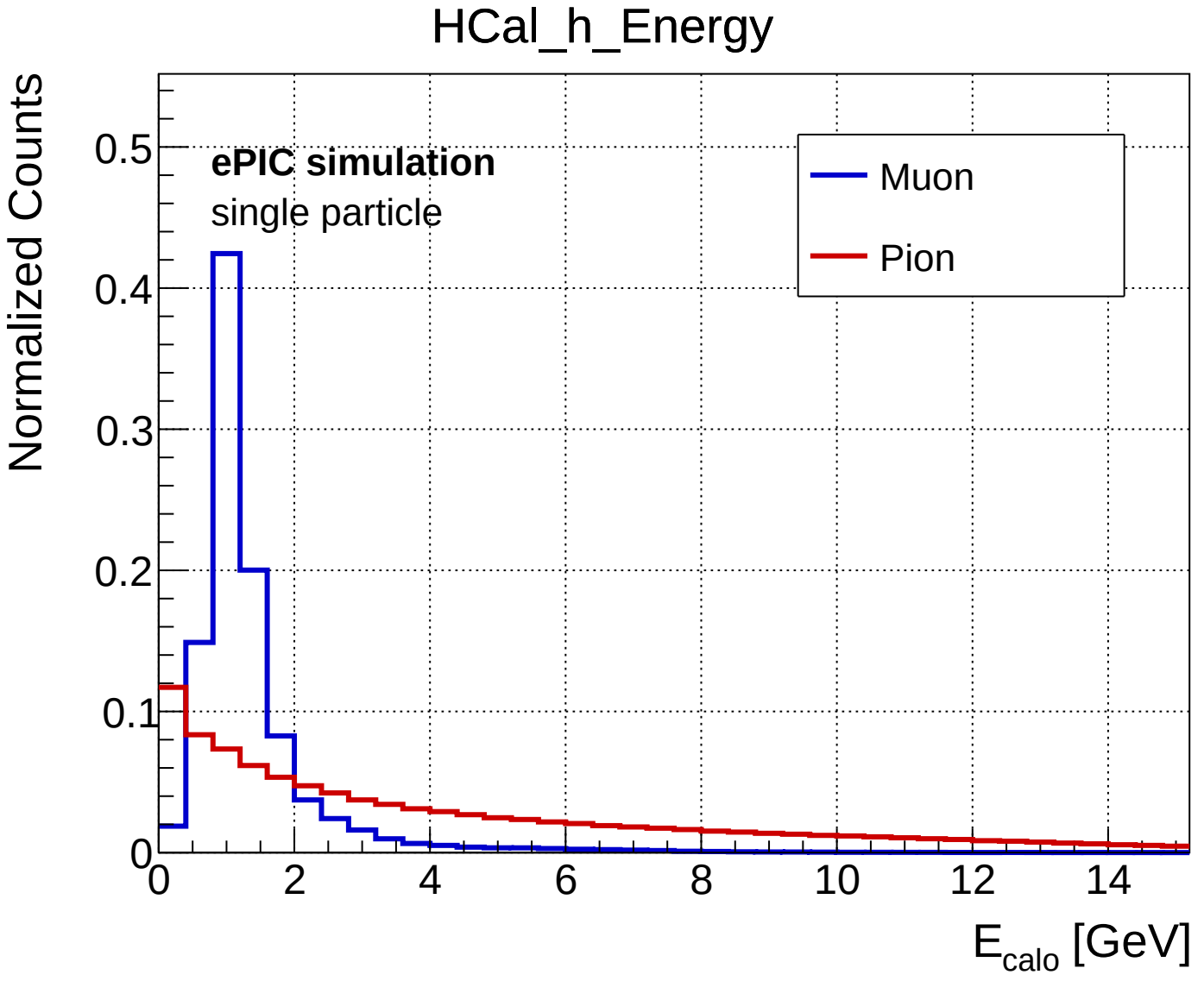

HCal_HitZ
Normalized Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
−2000
−1000
0
1000
2000
Z [mm]

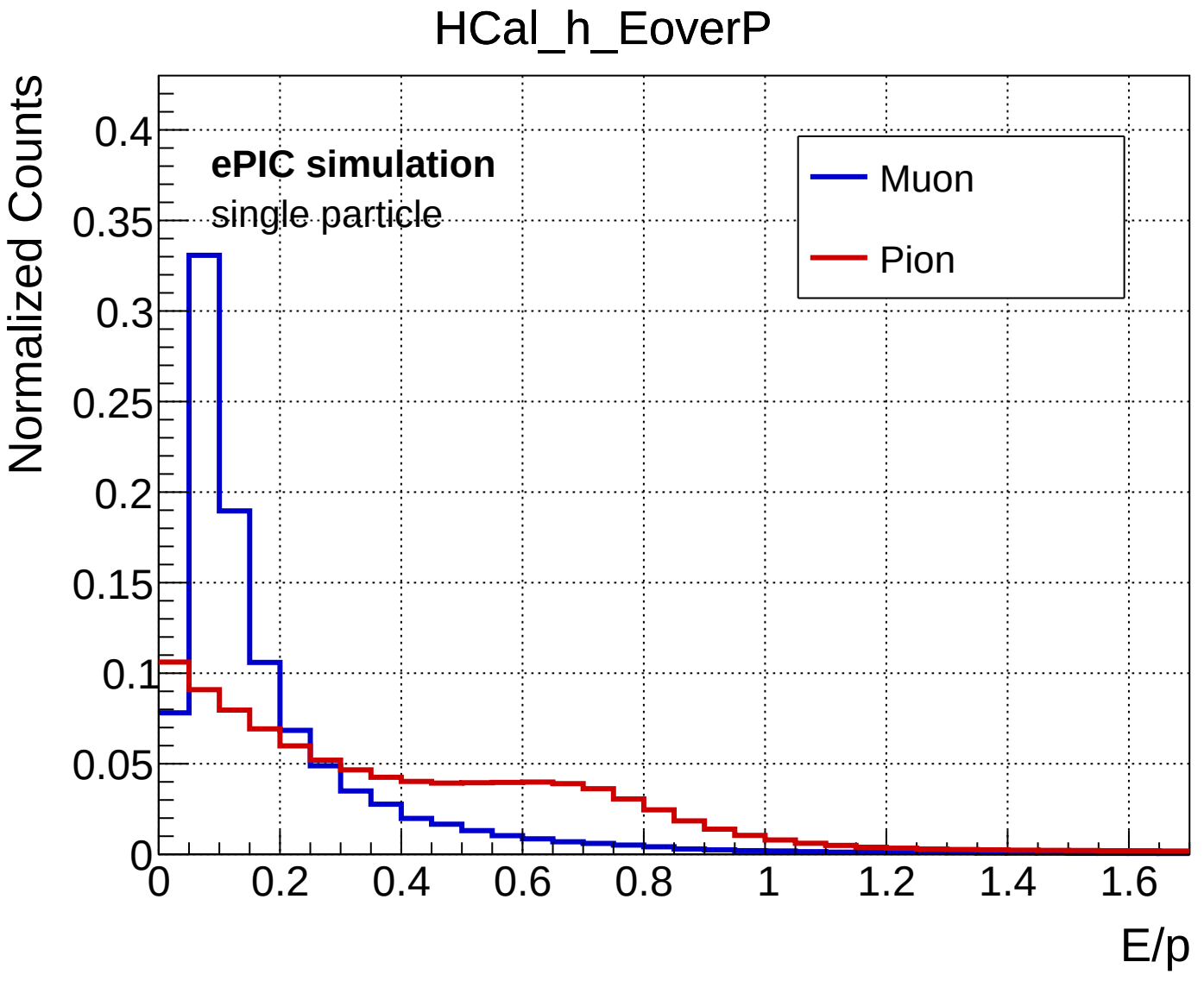

HCal_HitEnergy
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Energy [GeV]

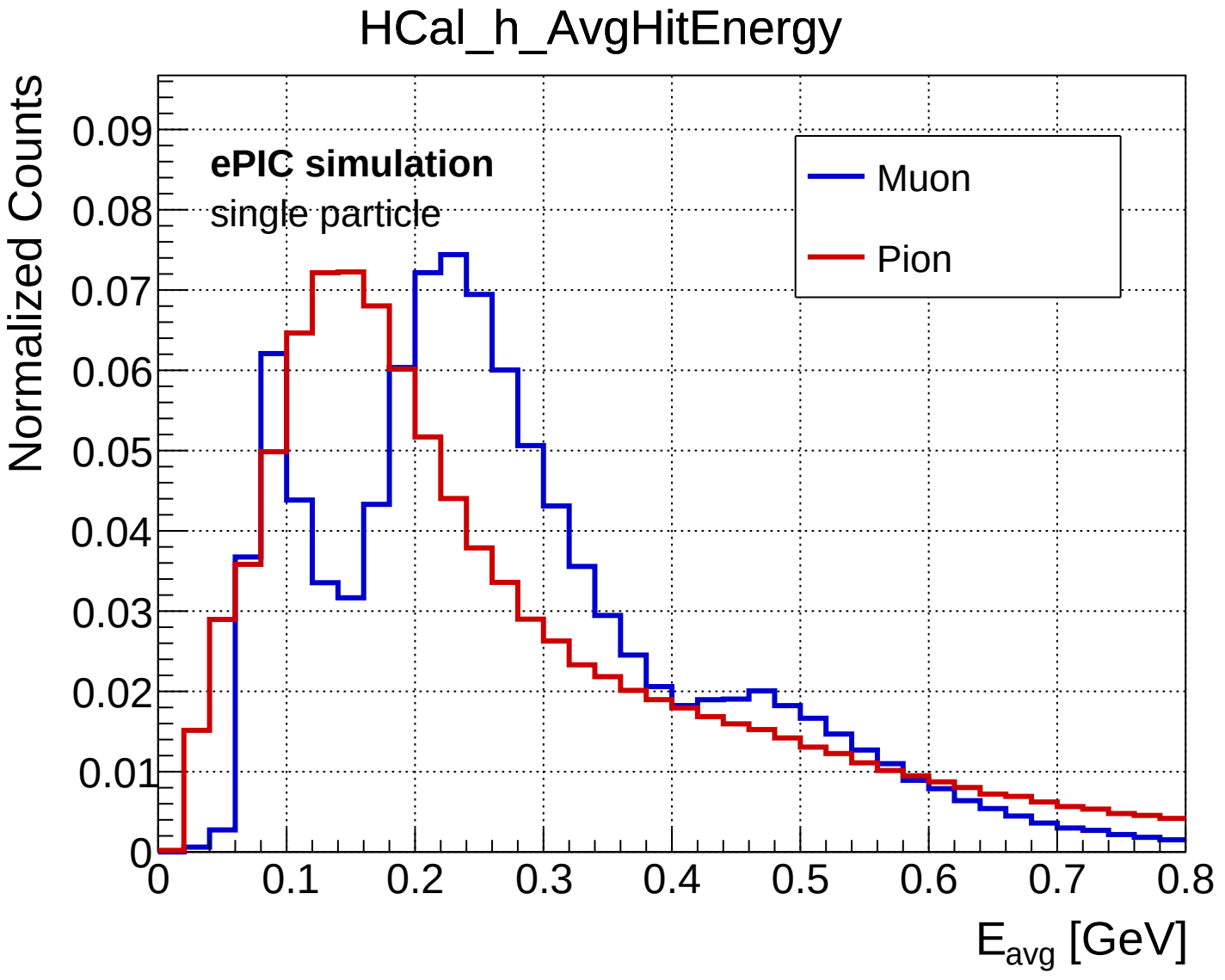

HCal_HitTime
Normalized Counts
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
5
10
15
20
25
30
35
40
45
Time [ns]

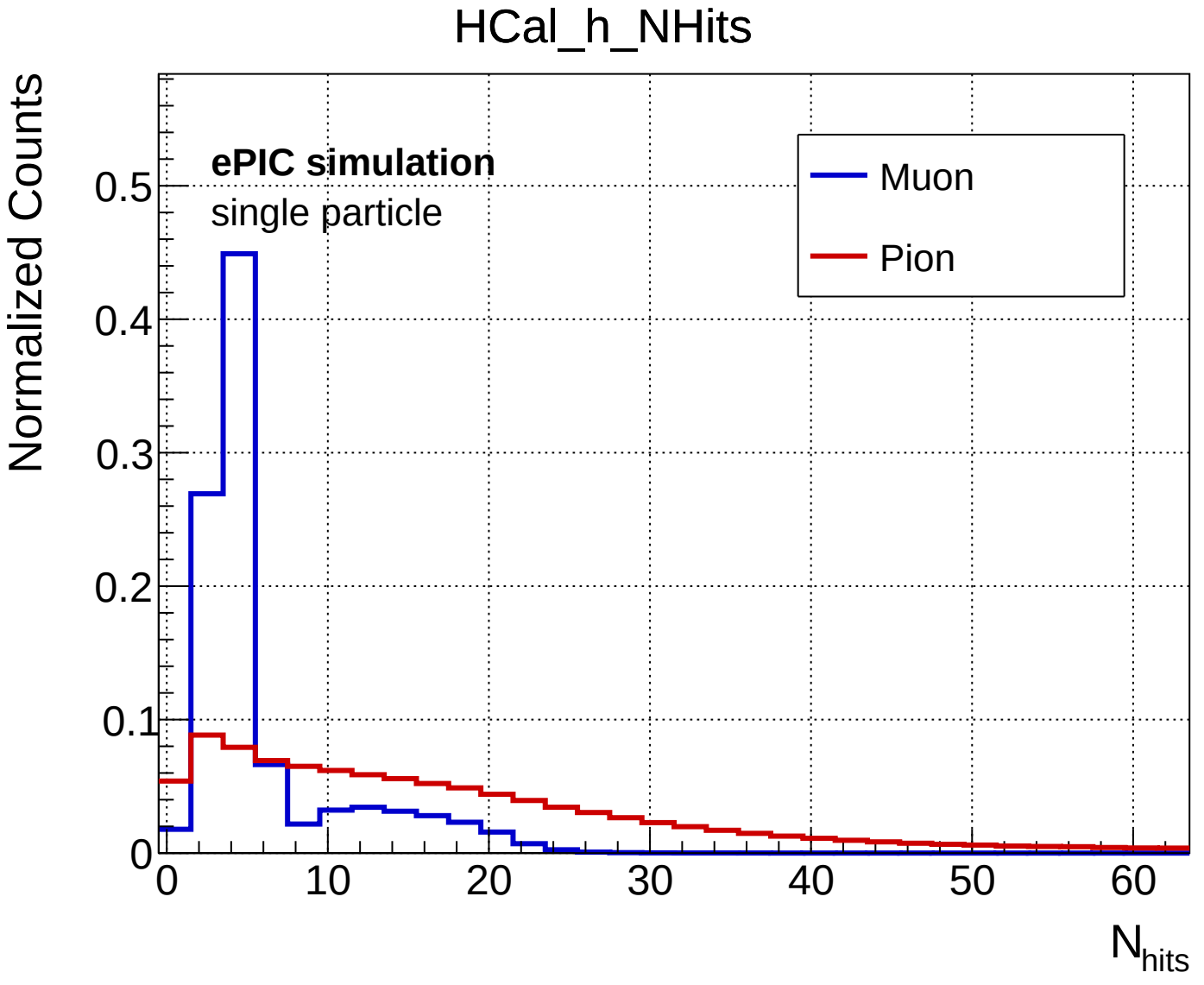

HCal_DeltaEta
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
–0.6
–0.4
–0.2
0
0.2
0.4
Δη

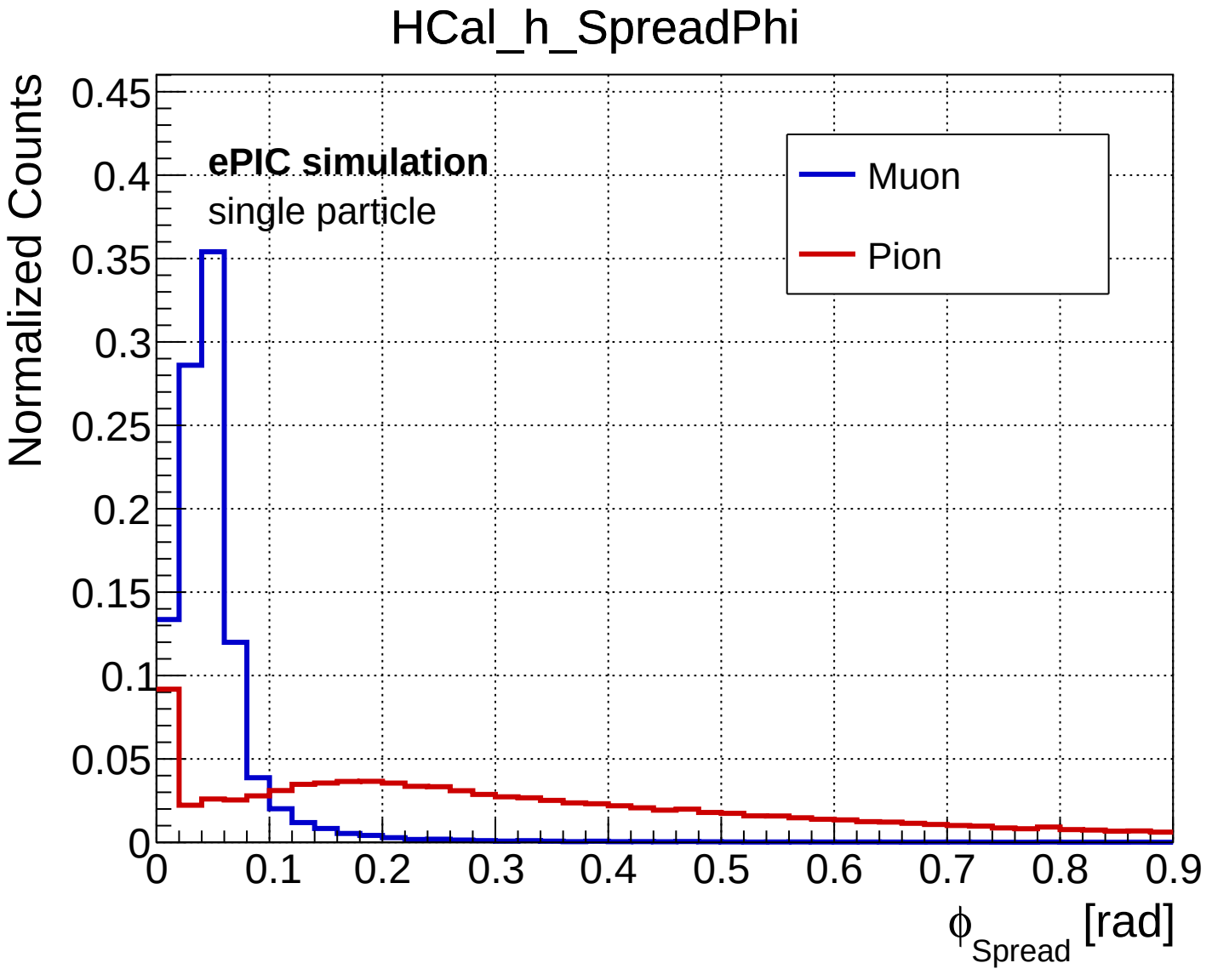

HCal_DeltaPhi
Normalized Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
–0.6
–0.4
–0.2
0
0.2
0.4
Δϕ

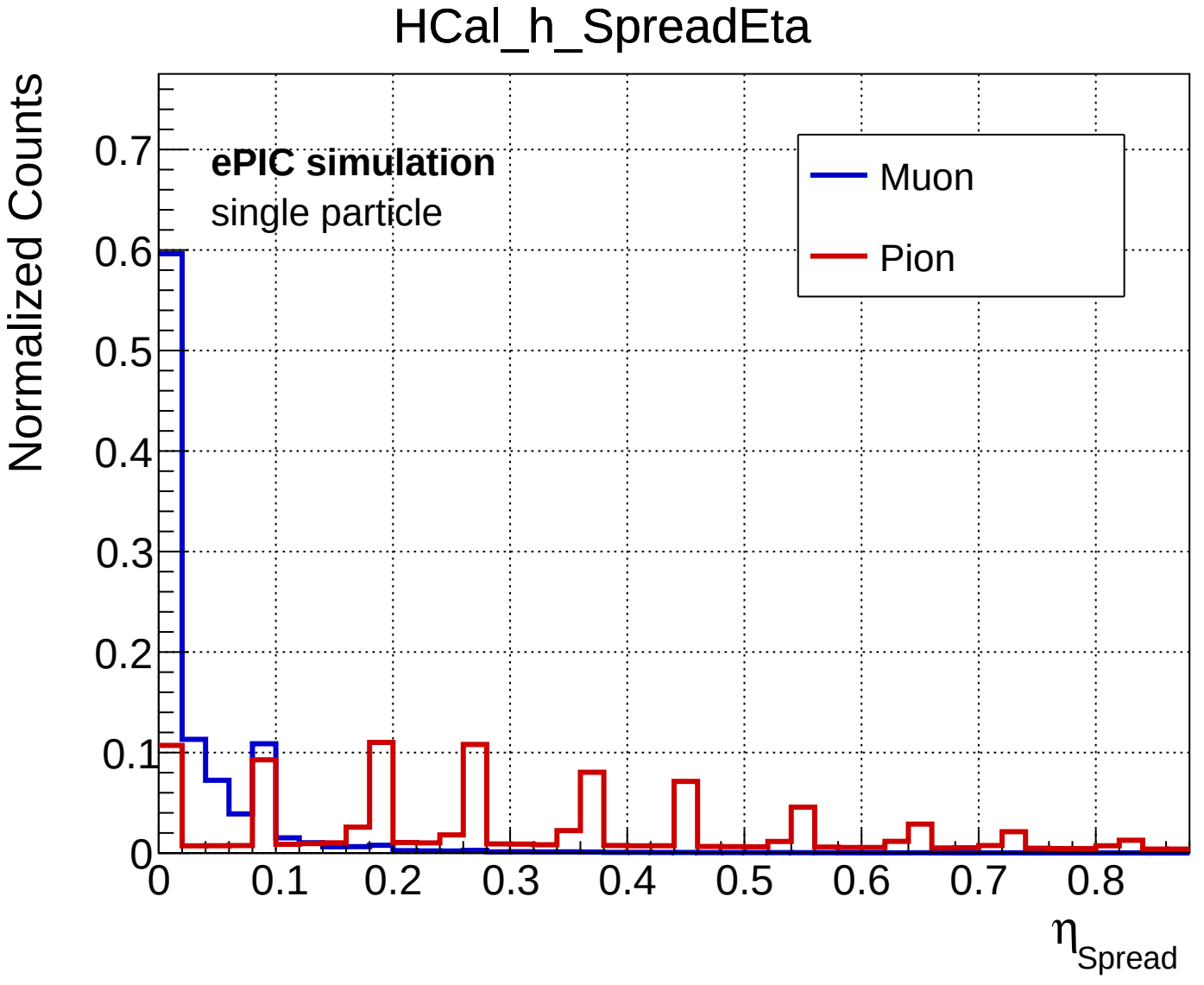

HCal_h_Energy
Normalized Counts
0.5
0.4
0.3
0.2
0.1
0
ePIC simulation
single particle
Muon
Pion
0
2
4
6
8
10
12
14
E_calo [GeV]

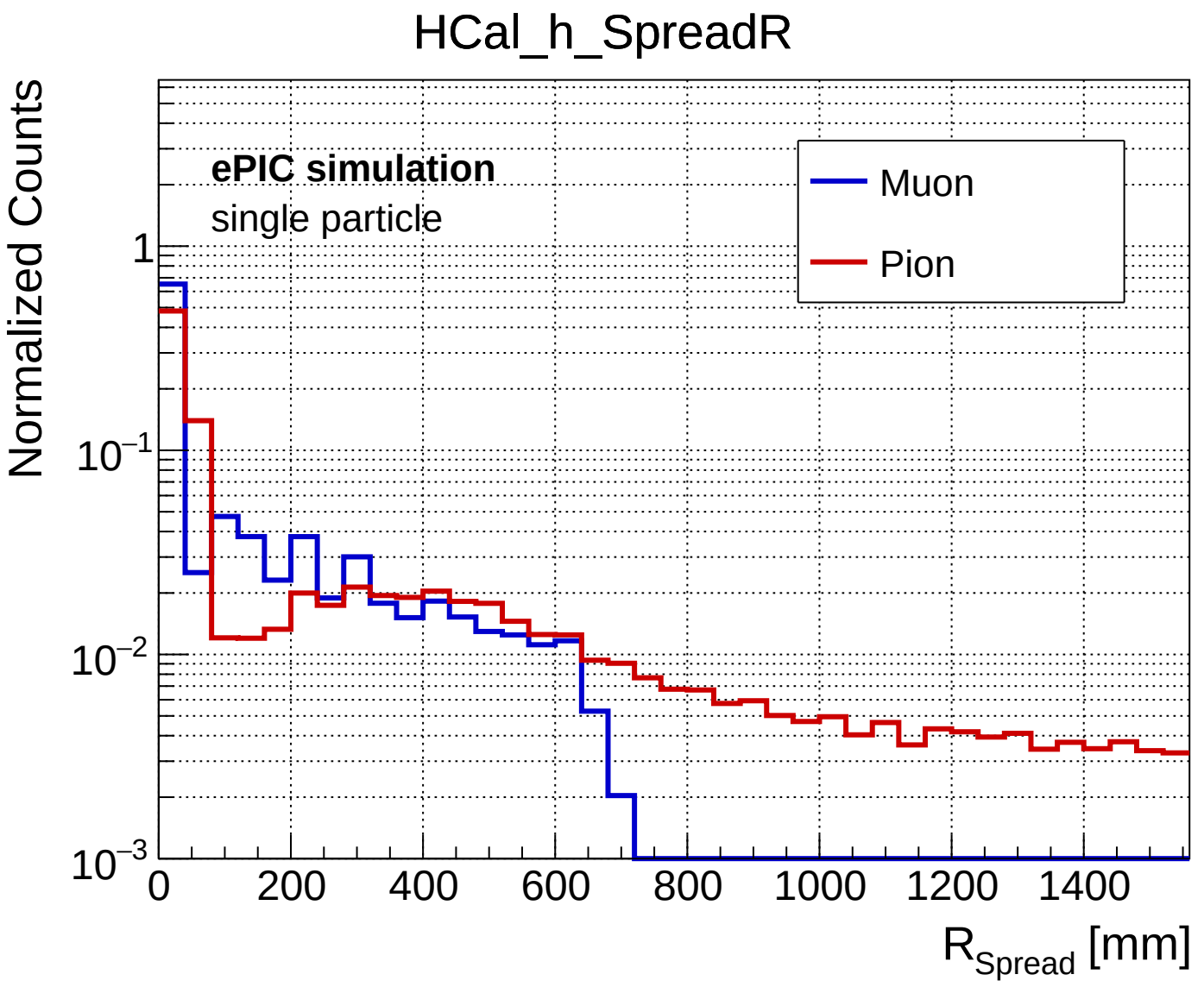

HCal_h_EoverP
ePIC simulation
single particle
Muon
Pion
Normalized Counts
E/p
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6

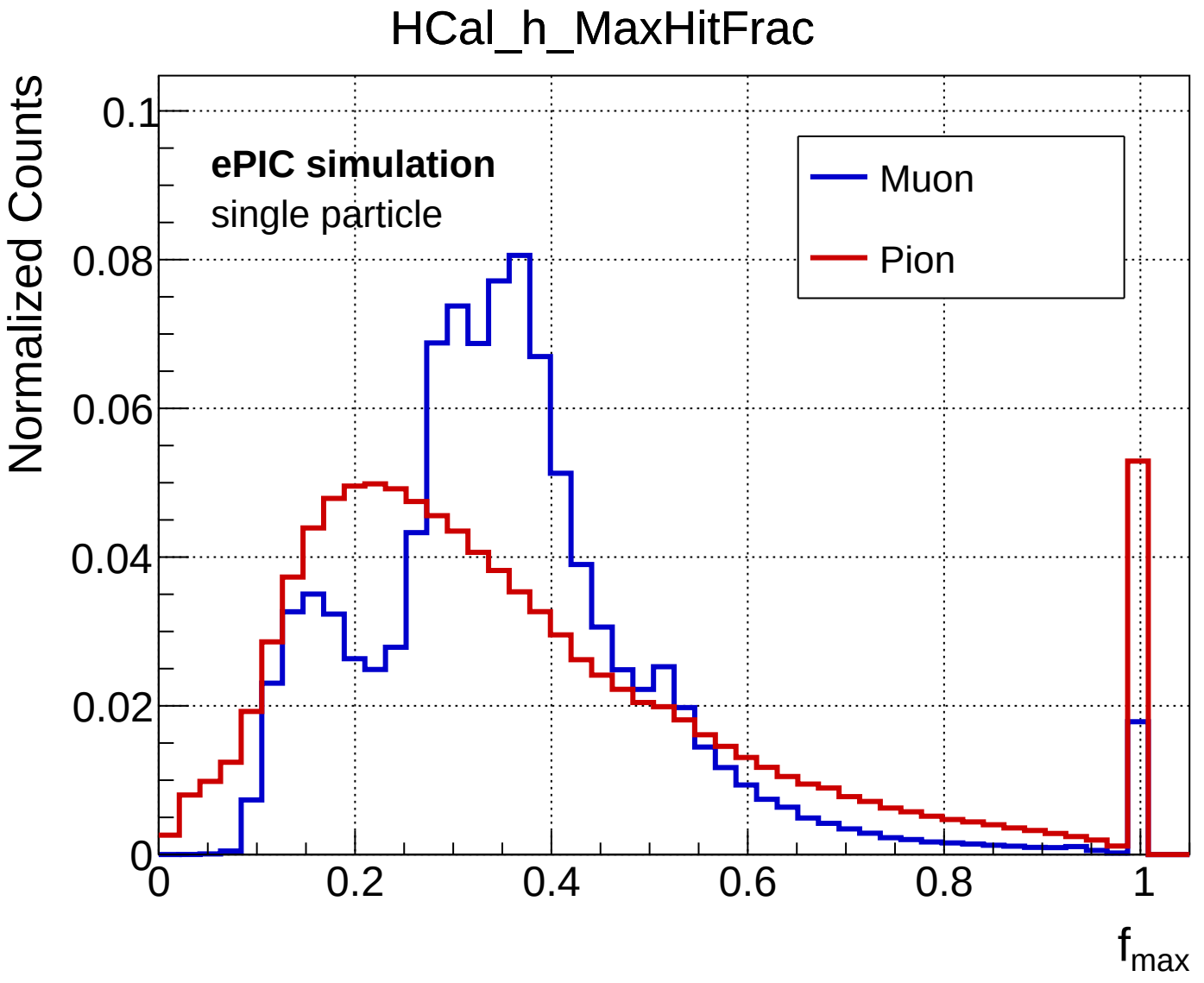

HCal_h_AvgHitEnergy
Normalized Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
E_avg [GeV]

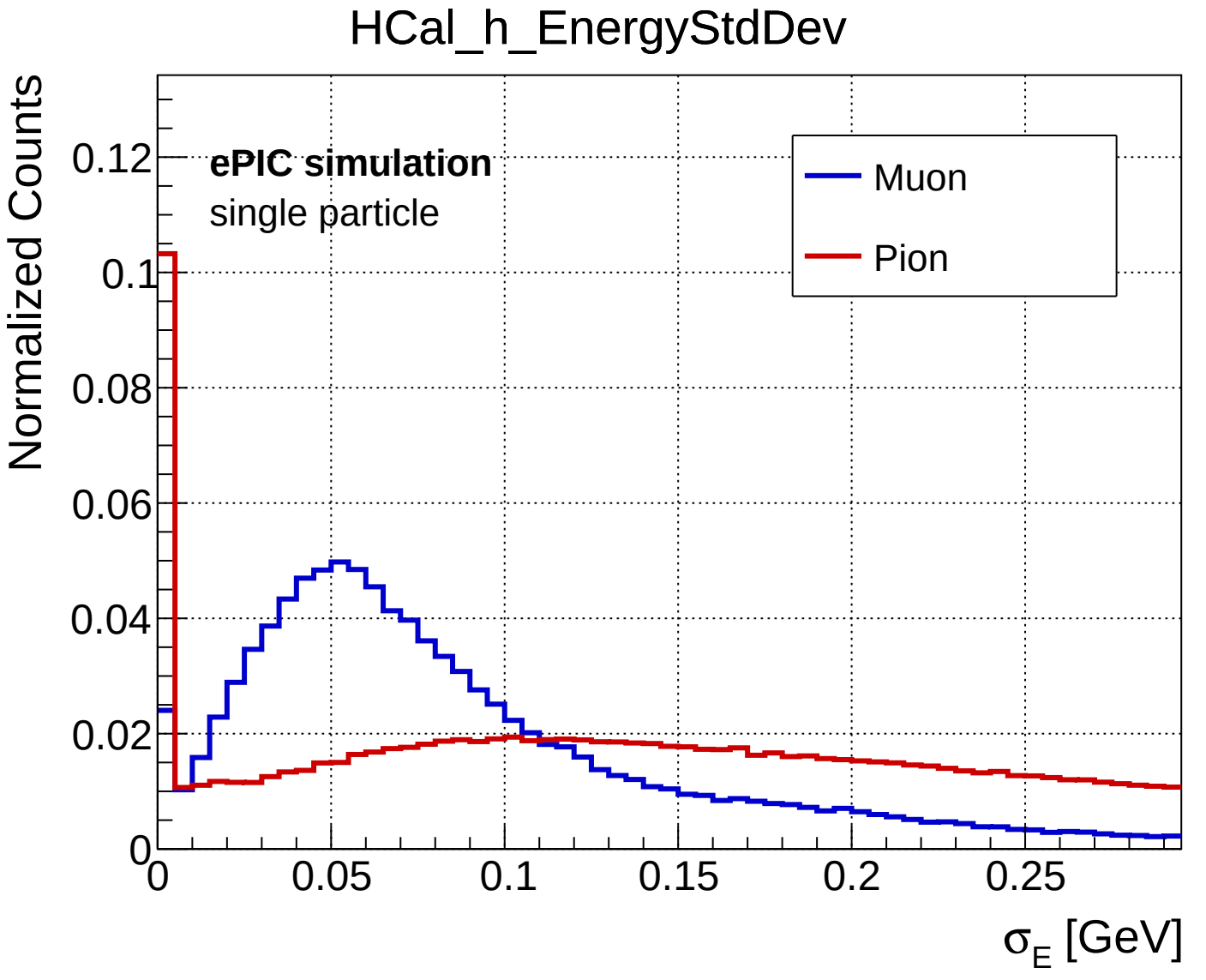

HCal_h_NHits
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50
60
N_hits

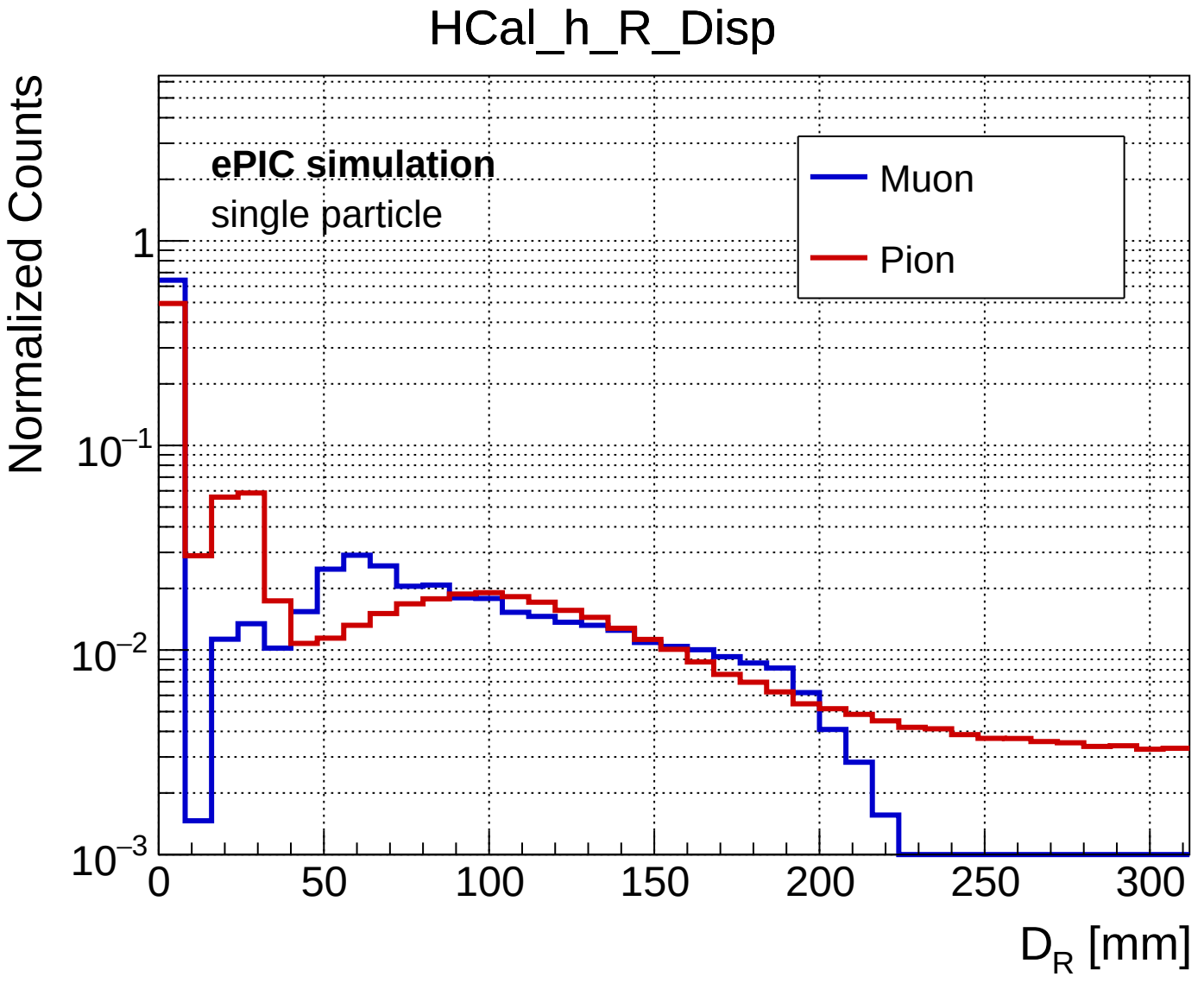

HCal_h_SpreadPhi
Normalized Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
ePIC simulation
single particle
Muon
Pion
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
ϕ_{Spread} [rad]

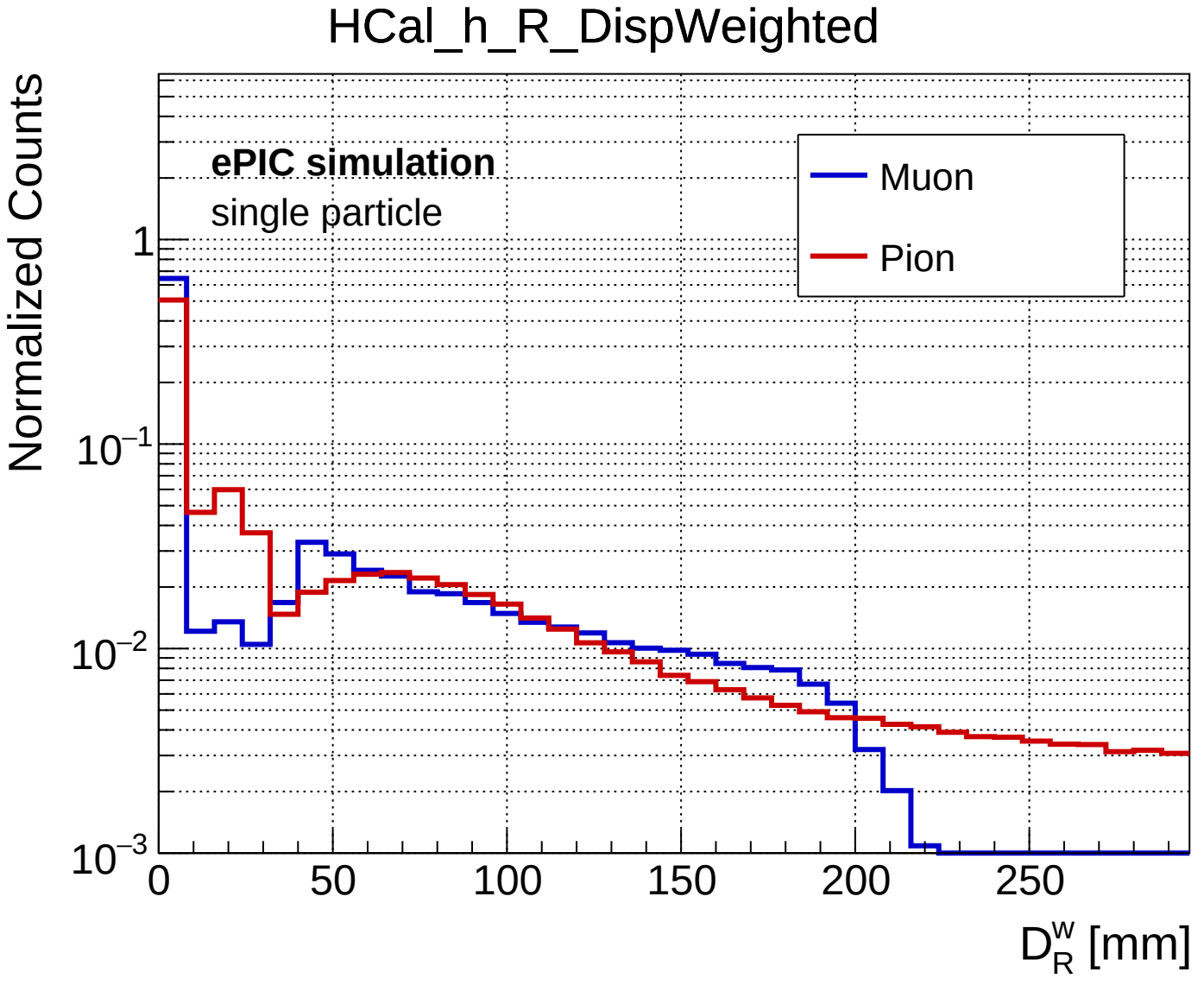

HCal_h_SpreadEta
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
η Spread

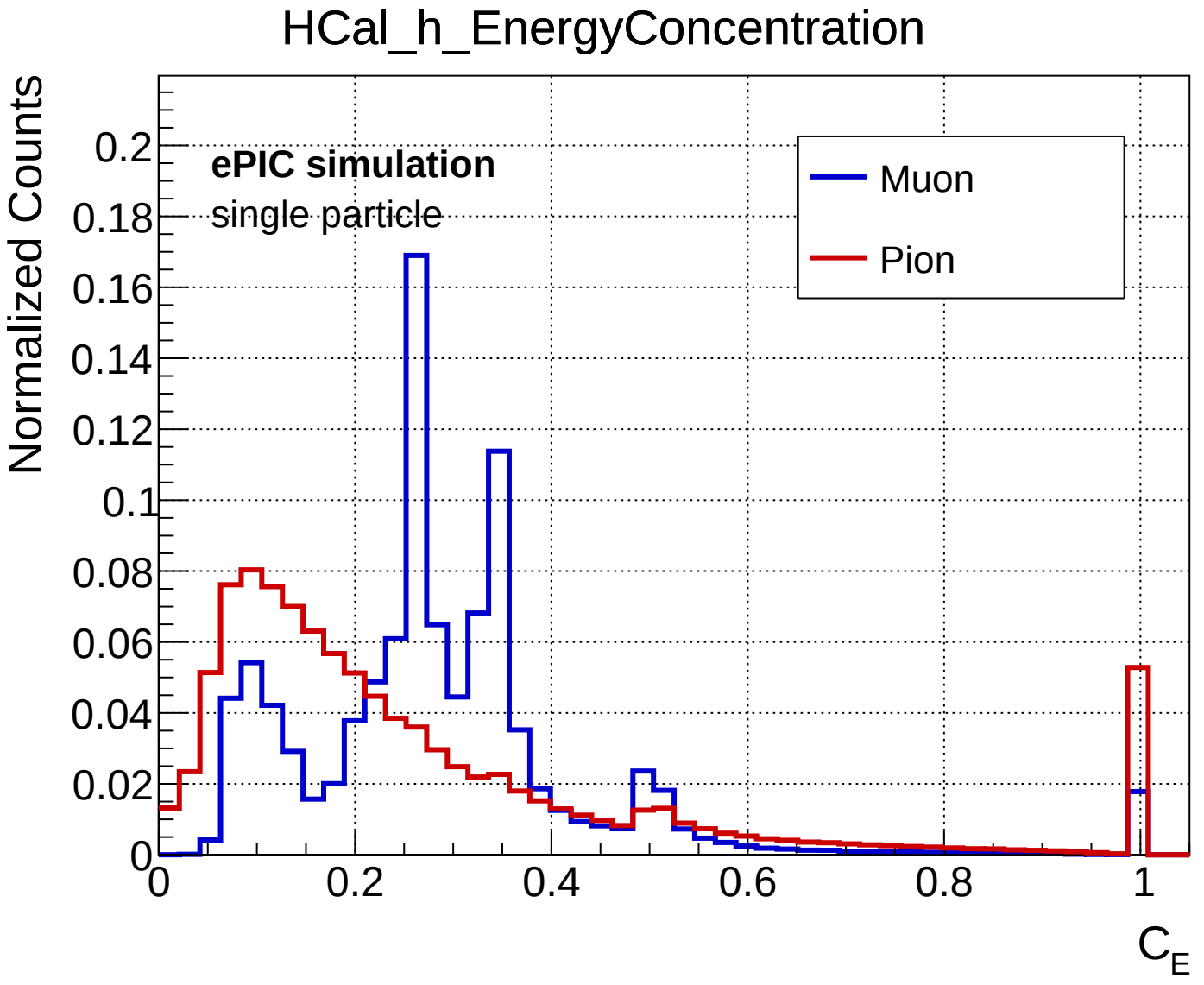

HCal_h_SpreadR
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
800
1000
1200
1400
R_{Spread} [mm]

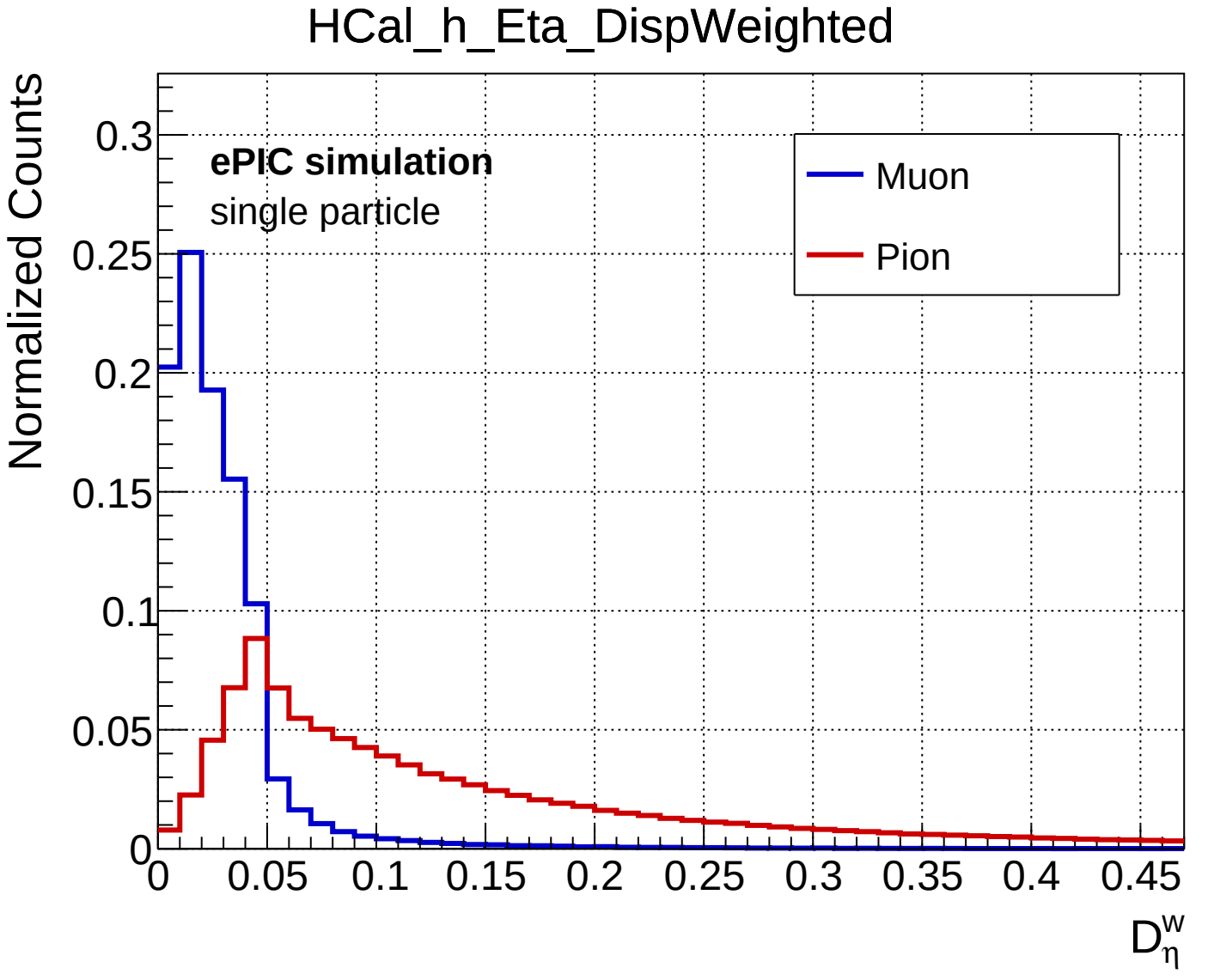

HCal_h_MaxHitFrac
Normalized Counts
0.1
0.08
0.06
0.04
0.02
0
0
0.2
0.4
0.6
0.8
1
f_{max}
ePIC simulation
single particle
Muon
Pion

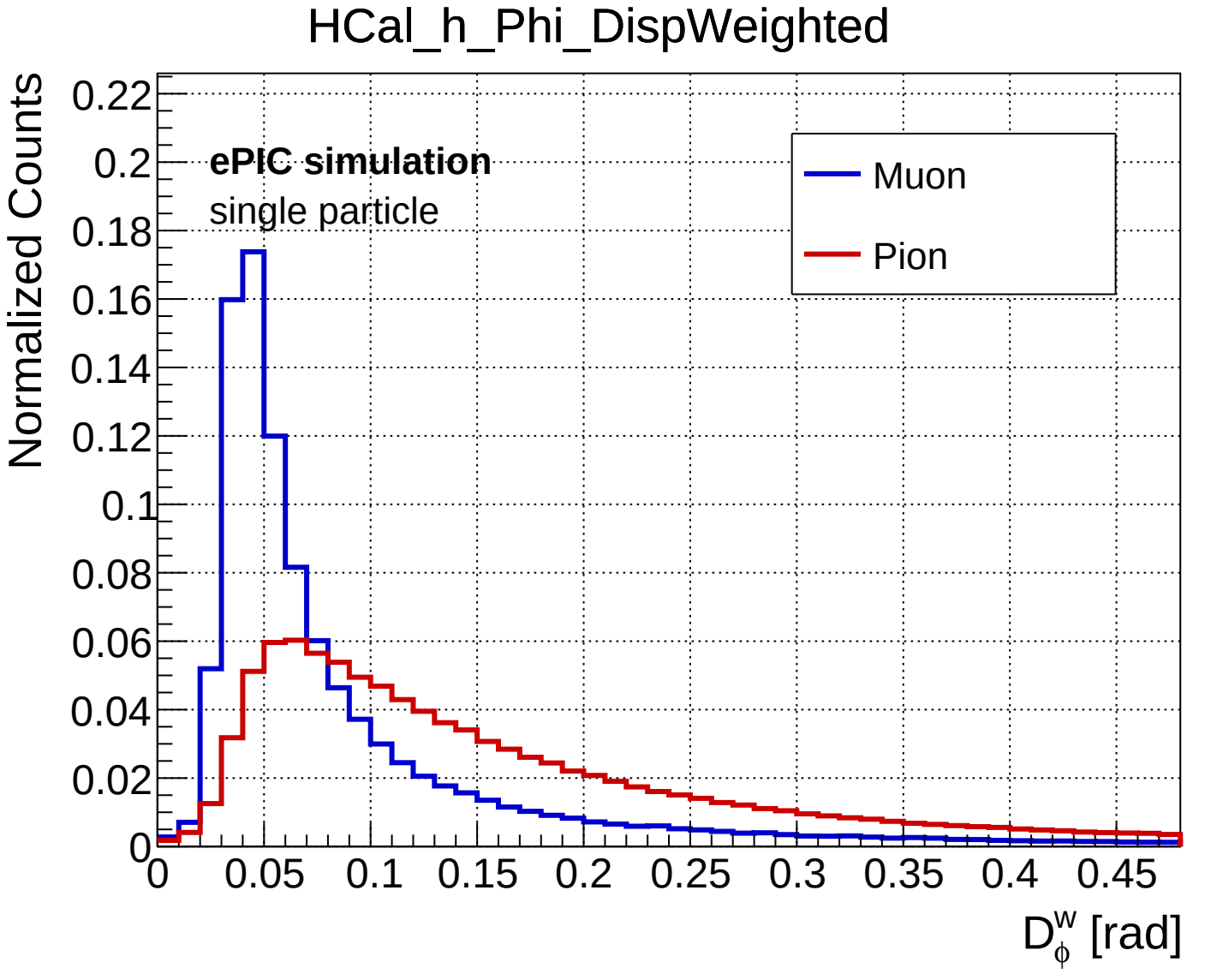

HCal_h_EnergyStdDev
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.05
0.1
0.15
0.2
0.25
σE [GeV]


HCal_h_R_Disp
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
300
D_R [mm]


HCal_h_R_DispWeighted
ePIC simulation
single particle
Muon
Pion
Normalized Counts
1
10^{-1}
10^{-2}
10^{-3}
0
50
100
150
200
250
D_R^w [mm]


HCal_h_EnergyConcentration
ePIC simulation
single particle
Muon
Pion
Normalized Counts
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.2
0.4
0.6
0.8
1
C_E


HCal_h_Eta_DispWeighted
Normalized Counts
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
D_η^w


HCal_h_Phi_DispWeighted
Normalized Counts
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
ePIC simulation
single particle
Muon
Pion
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
D_{ϕ}^{w} [rad]